Date: / / التاريخ: / /

Subject: Chap [1] Introduction موضوع الدرس:

Signal ← : مش شرط تعتمد على متغير واحد ممكن أكتر

زي بالفيديو مرتبط بـ Amp & time ، GPS عاوز x, y, z يقدر يحدد مكانه

Processing ← Filteration, Modulaton, Multiplix

- حاجه إيجابيه بتتعمل على ال signal

distortion ، Attenuation لا يعتبروا processing

system ← ; Hw or Sw مثل Filter, Microprocess

← باختصار، العمليه اللي بتتم على ال signal هي processing ومنها أقدر أحدد نوع ال system

filteration ← system ← Filter

Mux ← system ← Multiplexer

→ Classification of signal. ① According to characteristic

[1] Derminstic:

متوقع حدوثها وأقدر أعبر عنها رياضياً وكمان أعرف شكلها ex: sinwave

[2] Random:

غير متوقع حدوثها وصعب توقع شكلها أو قيمتها عند زمن معين ex Noise

② According to source:

[1] Multichannel:

signals مختلفه كل واحده جايه من source غير التانيه

→ temp sensor, pressure sensor Mux → one channel.

[2] Multidimentional:

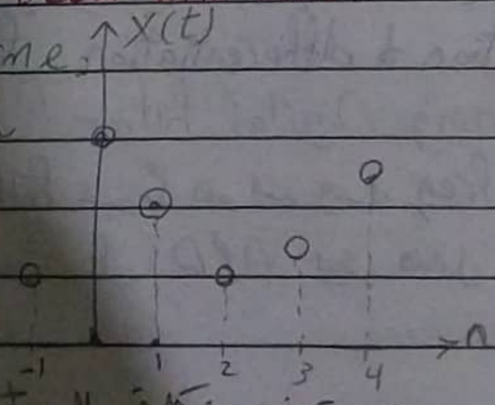
voice signal ← Amp & time ، gps x, y, z علشان أحدد مكانه

③ According to occur with time:

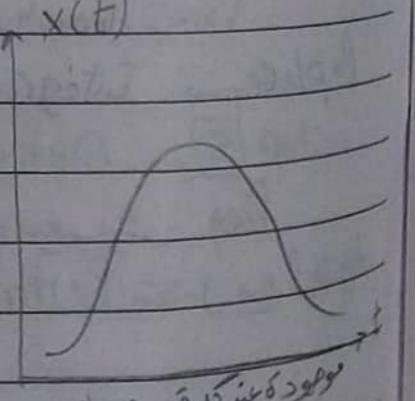
[1] Continous time:

[2] Discrete ~

samples عدد (n)



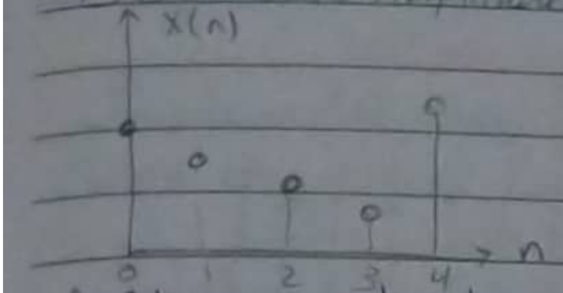
مش متعرفه عند كل قيم ال t



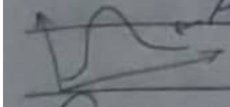
موجوده عند كل قيم من الزمن

(4) ACCording to Amplitude

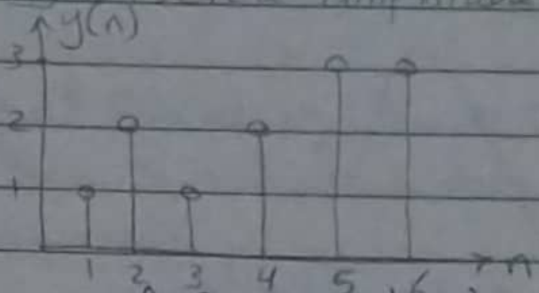
1 Continouse Amplitude



Amp الـ restrection مفيش معين على
كل sample واخده قيم غير معلوم ومش محددة
ودي أقدر أعتبرها Analog

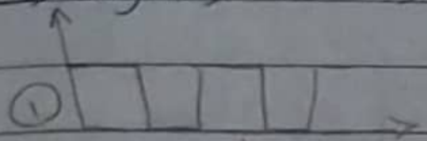


2 Discrete Amplitude



فيه restriction على الـ Amp
لازم كل sample تاخد قيم 1,2,3 ولو لا
بيتم تقريبها لأقرب رقم ليها ودي ليها شروط
بيحددها الـ quantizer

① → Analog signal → Continous in Amp & Continous oR Discret time
② → Digital ~ → ① Discrete in Amp & ~ ② ~ ~

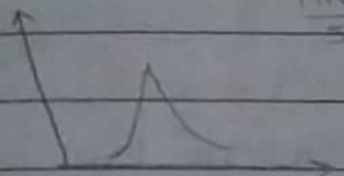


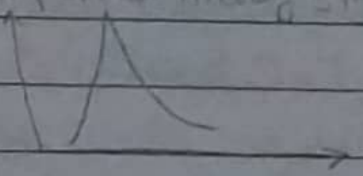
→ Types of signal processing

1 Analog signal Processing

Analog signal → A.S.P → Amplified Analog signal

Ex. Amplifier



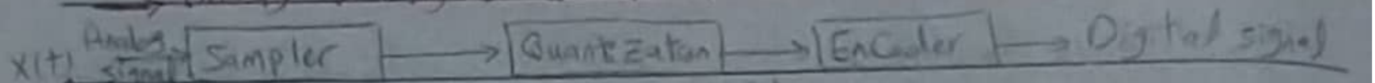


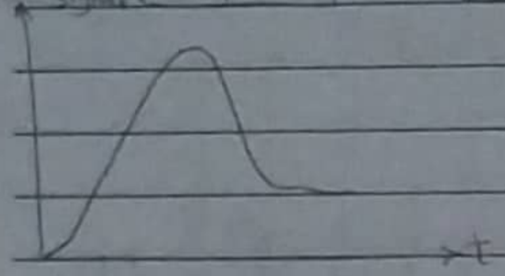
2 Digital signal processing

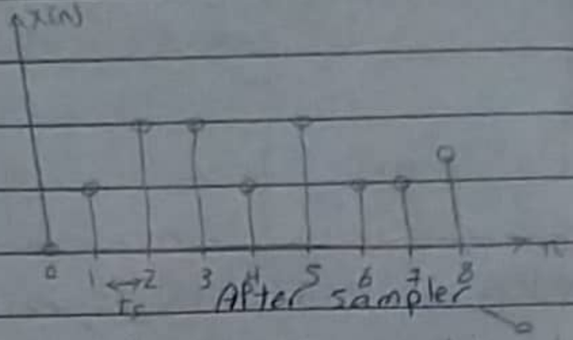
Analog signal → A.D.C → Digital signal → DSP → digital → D.A.C → Analog signal

① storage ② noise effect is law ③ Encryption
تشفير

→ Analog to Digital Conversion







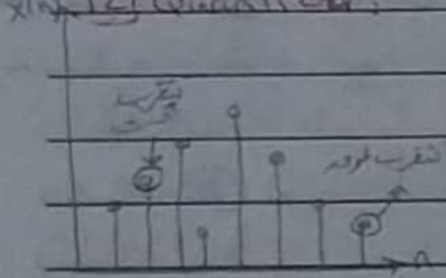
Input Analog signal   After sampler

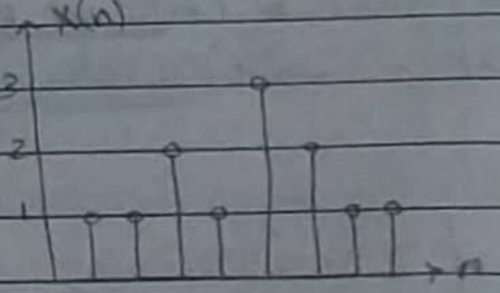
- عبارة عن switch بيقطع الـ signal يقفل ويوصل لفترة زمنية معينة Ts

1- ممكن اعمل Transistor يعمل automatic switch Continuous Amp Discrete time

x-axis → n sample x(n) o/p which I/p for Quantizer

[2] Quantizer:





- بيعمل عدد معين من الـ levels وبياخد أقرب كل قيمة لأقرب level طبقًا الـ Threshold

→ Discrete time to Discrete Amp

[3] Encoder:

بيعمل كل level الـ Quantization بكود معين له

No of bit in Code = 3 $2^3 = 8$

- أول sample عند level 0 ← كود 000 ثم 100 وهكذا

- طبعًا كل system بيحدد عاوز كام level وهيعبر عن الـ sample بكام bit طبقًا للظروف بتاعه

→ Why does the sinewave is of vital importance in signal processing ?

→ Fundamental principle of signal processing (Fourier transform) Called any signal Composed from Sum of sinusoidals of varying amplitude & frequency

$x(t) = A\cos(2\pi f t + \theta)$

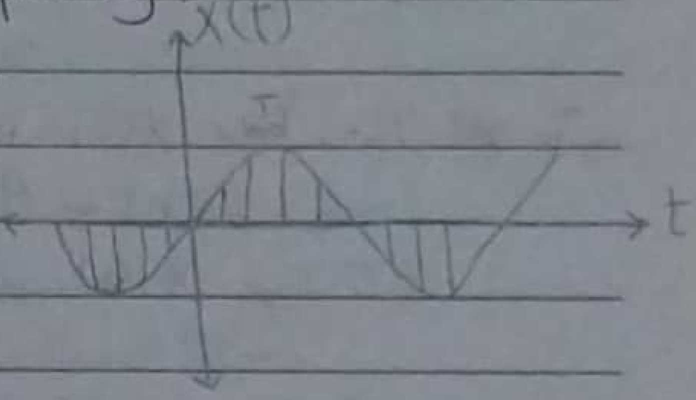
$\Omega = 2\pi f \qquad f = 1/T_p$

$x(t) = A\cos(\Omega t + \theta)$

$t = nT$

$\rightarrow x(n) = A\cos(\Omega n T + \theta)$

$= A\cos(\omega n + \theta)$



$f = \frac{F}{F_s} = FT$ → Just Counter

→ Sampling theram :

signal with bandwidth f_0 and sampling freq F_s

$F_s > 2f_0$ → Nyquist rate

[illegible] aliasing ← إن عملية ال sampling لازم تتم بأقل freq $(2f_0)$

T_p → periodic time (fundamental period)

(بعد تكرار ال cycle الواحدة (الإشارة تبدأ تكرر نفسها)

$x(t) = x(t + t_p)$ → ده ال periodicity

for Continuous time signal

→ Prove that : $x(t) = x(t + t_p)$

$x(t) = A\sin(2\pi f t + \theta)$ replace t by $t + t_p$

$x(t + t_p) = A\sin(2\pi f(t + t_p) + \theta)$

$x(t + t_p) = A\sin(2\pi f t + 2\pi f t_p + \theta)$ $f = \frac{1}{t_p} \rightarrow f t_p = 1$

$= A\sin(2\pi f t) + A\sin(\cancel{2\pi} + \theta)$

$A\sin(\theta)$

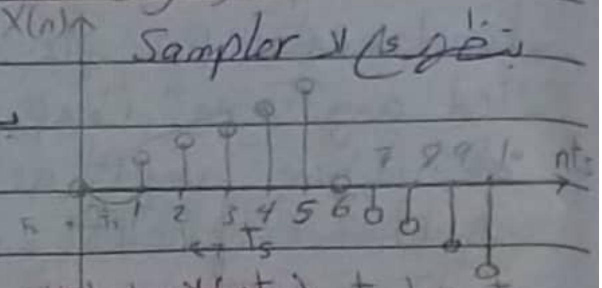
$\longrightarrow x(t+t_p) = A\sin(2\pi f t + \theta)$

$\therefore x(t+t_p) = x(t)$ Periodic signal

[2] Periodicity for Discrete time signal.

Ts مرة Times nt_s هنا بيعمل ا ـ

بيمركز sample



→ To get x(n) from x(t) replace ~~x(t) by x(nts)~~ t by ts

$\longrightarrow x(n) = A\sin(2\pi f(nt_s) + \theta)$, $T_s = 1/f_s$

$\rightarrow x(n) = A\sin(2\pi \frac{f}{f_s} n + \theta)$ $\quad \frac{f}{f_s} = F$

$\longrightarrow x(n) = A\sin(2\pi F n + \theta)$

where $F = \frac{f}{f_s} = \frac{\text{cycle/sec}}{\text{sample/sec}} = \frac{1/T_p}{1/T_s} = \frac{T_s}{T_p}$ (cycle/sample)

Digital freq

$f_s \rightarrow$ No of samples per certain time (sampling freq)

(Fundemental period) Np هيكون بدل Tp

→ prove that $x(n) = x(n+N_p)$

$x(n) = A\sin(2\pi f nt_s + \theta)$

$= A\sin(2\pi F n + \theta)$

replace n by $n + N_p$

$\longrightarrow x(n) = A\sin(2\pi F(n+N_p) + \theta)$

$x(n) = A\sin(2\pi F n + \underline{2\pi F N_p} + \theta)$

$\longrightarrow x(n) = A\sin(2\pi F n + \theta)$ $\neq$

if $F N_p = K$ as K is integer No

$F = \frac{K}{N_p} = \frac{\text{integer}}{\text{integer}}$

لازم ده كله يتحقق

علشان نقدر أقول انها periodical

Date / / التاريخ / /

Subject: sheet (1) $F_s = 1/T_s$ موضوع الدرس

check periodicity

$\boxed{1}$ a) $\cos(0.01 n\pi)$ → discrete

① → $A\cos(2\pi F n + \theta) = \cos(0.01 n \pi)$

$2\cancel{\pi} F \cancel{n} = 0.01 \cancel{n} \cancel{\pi}$ $2F = 0.01$

$F = \frac{0.01}{2} = \frac{1}{200} = \frac{k}{N_p}$ → ①

F → rational number

② check $X(n) = X(n + N_p)$

$X(n+N_p) = \cos(0.01\pi(n + \overset{\to N_p}{200}))$

$= \cos(0.01\pi n + \cancel{2\pi})$

$= \cos(0.01\pi n)$

ال Cos لما اضيف عليها 360 هترجع زي ما هي تاني

$\therefore$ $X(n) = X(n + N_p)$ periodic

b) $\cos(n \frac{30}{105}\pi)$ → discrete

$\cos(n\frac{30}{105}\pi) = A\cos(2\pi F n + \theta)$

$2\cancel{\pi} F \cancel{n} = \cancel{n} \frac{30}{105} \cancel{\pi}$ $2F = \frac{30}{105}$

$F = \frac{30}{210} = \frac{1}{7} = \frac{k}{N_p}$ rational ①

check ② $X(n) = X(n + N_p)$

$X(n+N_p) = \cos(\frac{30}{105}\pi n + \frac{30}{105}\pi \overset{7}{\cancel{N_p}})$

$= \cos(\frac{30\pi n}{105} + \cancel{2\pi})$

$\therefore X(n) = X(n + N_p)$ Periodic

Ex.(1.6): a) $x(t) = 3\cos\left(5t + \frac{\pi}{6}\right)$

a) prove periodicity:

$$x(t) = x(t + t_p)$$

$$x(t+t_p) = 3\cos\left(5(t+t_p) + \frac{\pi}{6}\right) \rightarrow ①$$

we have $\Longrightarrow$ $3\cos\left(5t + \frac{\pi}{6}\right) = A\cos(2\pi f t + \theta)$

$$2\pi f \cancel{t} = 5 \cancel{t} \qquad 2\pi f = 5$$

$$f = \frac{5}{2\pi}\ Hz \qquad t = \frac{2\pi}{5}\ sec$$

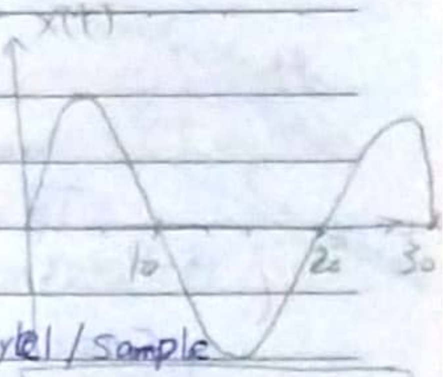
at ① $\Longrightarrow$ $x(t+t_p) = 3\cos\left(5t + \cancel{5} * \frac{2\pi}{\cancel{5}} + \frac{\pi}{6}\right)$

$$x(t+t_p) = 3\cos\left(5t + \frac{\pi}{6} + \cancel{2\pi}\right)$$

$$\Longrightarrow x(t) = x(t+t_p)$$

(1.7) $x(t) = 3\sin(\underline{100}\pi t)$ $2\pi f$

a) sketch The signal for [0, 30] msec



b) $F_s = 300$ Determine f and show It is periodic

F → Discrete time signal freq cycle / sample

f → Continous time freq (cycle/sec)

f_s → sampling freq

$T_p = \frac{1}{50} = 0.2$ sec

$T_p = 20$ msec

$F = \frac{f}{f_s} = \frac{50}{300} = \frac{1}{6}$ Cycle / sample

عشان (cycle) اطلع ع 6 samples

$F = \frac{1}{6}$ rational no $\frac{k}{N_p}$ $N_p = 6$

c) Compute sample values of x(n), sketch x(n) what is The period?

$x(n) = 3\sin(100\pi n t_s)$ $f_s = 300$ $t_s = \frac{1}{300}$

$x(n) = 3\sin(\frac{100}{300}\pi n)$

$x(n) = 3\sin(\frac{n\pi}{3}) \longrightarrow ①$

$x(n+N_p) = 3\sin(\frac{n\pi + N_p\pi}{3})$

$= 3\sin(\frac{n\pi}{3} + \frac{2\cancel{6}\pi}{\cancel{3}})$

$x(n+N_p) = 3\sin(\frac{n\pi}{3}) \rightarrow ②$

$x(0) = 0$ $\therefore x(n) = x(n+N_p)$ Periodic

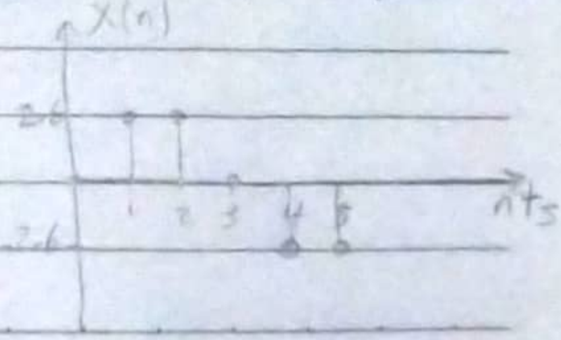
$x(1) = 3\sin(\frac{\pi}{3}) = 2.6$ * عاوز قيم ال sample من 0 ← 5

$x(2) = 3\sin(\frac{2\pi}{3}) = 2.6$

$x(3) = 3\sin(\pi) = 0$

$x(4) = 3\sin(\frac{4\pi}{3}) = -2.6$

$x(5) = 3\sin(\frac{5\pi}{3}) = -2.6$



$T_s = \frac{1}{f_s} = \frac{1}{300} = 3\frac{1}{3}$ msec

Date: / / التاريخ: / /

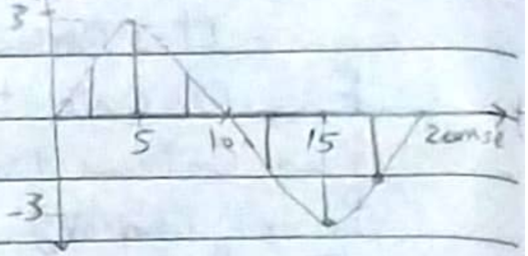
Subject: موضوع الدرس:

مش علينا الجزء ده

d) ← عاوز أشوف ال f_s الأقل الي تجيب ال peak

$100\pi t = 2\pi f t \qquad f = 50$

$T_p = \frac{1}{f} = \frac{1}{50} = 20\,\text{msec}$



ال peak هيجي عند 5msec

Sampe every $T_s = 1\,\text{msec}$

$T_s = 5\,\text{msec} \Rightarrow f_s = \frac{n}{5\,\text{msec}} = 200n$

أقل sampling ممكن أخده 200 samples

علشان ال peak توصل ل (3)

ده المطلوب

طريقة الشرح نفسها

$X(n) = 3\sin(2\pi F n)$

$1 = \sin(2\pi F n)$

$2\pi F n = \sin^{-1}(1) = 90 = \frac{\pi}{2}$

$F = \frac{1}{4n} = \frac{f}{f_s} \qquad f_s = 4n * f$

$= 4n * 50 = 200n$

200 sample/sec

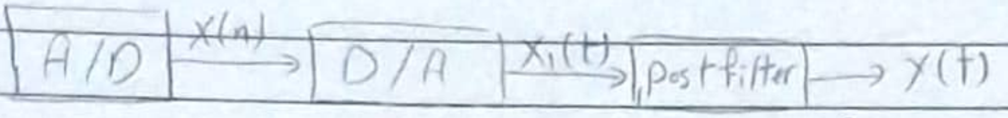
T_p → periodic time (fundemental period)

N_p → fundemental period sample.

T_s → sampling time

(EX) 1.9: Consider a simple signal processing ~~signal~~
The sampling periods of A/D & D/A Converts $T_1 = 5ms$
$T_2 = 1ms$ Determine the o/p if I/p is

$$x(t) = 3\cos(100\pi t) + 2\sin(250\pi t)$$



كده هتقدم cut of freq هيعدي signal ويرمي ال noise

$f_1 = \frac{100}{2} = 50 Hz$ $f_2 = \frac{250}{2} = 125 Hz$

~~$f = 250$~~ $f = 125 Hz$ علشان تعدي الإشارة

→ x(n) by each t with nt_s

$$x(n) = 3\cos(100\pi n t_s) + 2\sin(250\pi n t_s)$$

$t_1 = 5 msec$

$$= 3\cos(100\pi n * 5*10^{-3}) + 2\sin(250\pi n * 5*10^{-3}$$

$$x(n) = 3\cos(0.5\pi n) + 2\sin(1.25\pi n).$$

→ x(t) by each n by $\frac{t}{t_s}$ where $T = nT_s$ $n = \frac{T}{t_s}$

$$x(n) = 3\cos(0.5\pi \frac{t}{1*10^{-3}}) + 2\sin(1.25\pi \frac{t}{1*10^{-3}})$$

$2\pi f_1 t = \frac{0.5\pi t}{10^{-3}}$ $2\pi f_2 t = \frac{1.25\pi t}{10^{-3}}$

$f_1 = 250 Hz$ $f_2 =$ ~~250Hz~~ $625 Hz$

$\boxed{y(t) = Zero}$ يعني مفيش signal خرج

CamScanner

Date: / / التاريخ: / /

Subject: Chapter (2) موضوع الدرس:

Part (1) Discrete Fourier Transform (DFT)

→ To get spectrum of signal There are several ways.

1. Discrete Fourier series :- for periodic & discrete signal
2. Fourier series :- for periodic & Continous signal
3. Fourier Transform :- for non-periodic & Continous signal or discret

in this part we will work with Fourier Transform →

1 Discrete Fourier transform (DFT) :-

$X(n) \rightarrow X(K)$ ⟵ تحويل الـ signal الـ Aperiodic إلى discrete ⟵

$$\boxed{X(k) = \sum_{n=0}^{N-1} X(n)\, e^{-Jnk\frac{2\pi}{N}}}$$

n → sample No

K → harmonic No

N → Total No of samples

example :-

1 Find X(K) or spectrum of X(n)

for $X(n) = \{0, 1, 1, 0\}$ Amp for samples

$\Rightarrow X(0) = 0 \;,\; X(1) = 1 \;,\; X(2) = 1 \;,\; X(3) = 0$

$$\Rightarrow X(K) = \sum_{n=0}^{N-1} X(n)\, e^{-Jnk\frac{2\pi}{N}} \qquad N = 4 \text{ sample}$$

$$\therefore X(K) = \sum_{n=0}^{3} X(n)\, e^{-Jnk\frac{2\pi}{4}} = \sum_{0}^{3} X(n)\, e^{-Jnk\frac{\pi}{2}}$$

$$\Rightarrow X(K) = \overset{0}{\cancel{X(0)}}\overset{0}{e} + \overset{1}{\cancel{X(1)}}\, e^{-JK\frac{\pi}{2}} + \overset{1}{\cancel{X(2)}}\, e^{-2JK\frac{\pi}{2}} + \overset{0}{\cancel{X(3)}}\, e^{-3JK\frac{\pi}{2}}$$

$$\Rightarrow X(K) = \overset{1}{\cancel{X(1)}}\, e^{-JK\frac{\pi}{2}} + e^{-2JK\frac{\pi}{2}}$$

خاصية الـ Duality تقول لو عندي signal في الـ Time domain وهو Aperiodic ⟵ التحويل لـ الـ freq domain بتكون periodic

وبالتالي عندي 4 harmonics هيبدوا يتكرروا بعد الـ (4)

$$\therefore X(K) = e^{-JK\frac{\pi}{2}} + e^{-JK\pi}$$

Date : / /

Subject :

at $K=0$:- $X(0) = e^{0} + e^{0} = 2$

at $K=1$:- $X(1) = e^{-J\frac{\pi}{2}} + e^{-4J\frac{\pi}{2}}$

$= -J - 1$

$X(1) = -1 - J$

at $K=2$:-

$X(2) = e^{-J \times 2 * \frac{\pi}{2}} + e^{-J * 2 * \pi}$

$= e^{-J\pi} + e^{-J2\pi}$

$= -1 + 1 = 0$

at $K=3$:- مضاعفات $\frac{\pi}{2}$

$X(3) = e^{-J * \frac{3\pi}{2}} + e^{-J * 3 * \pi}$ مضاعفات π

$= J - 1$

$X(3) = -1 + J$

*hint :-

$e^{\pm J\theta} = \cos\theta \pm J\sin\theta$

special Cases :-

① $e^{\pm Jm\pi}$ مضاعفات π

m even No $= \pm 1$ ①

m odd No $= \mp 1$ (-1)

② $e^{\pm Jm\pi/2}$ مضاعفات $\frac{\pi}{2}$

$m = 1, 5, 9, \ldots$ $= \pm J$

$m = 3, 7, 11, \ldots$ $= \mp J$

* Conjugate For X(1) كل قيم لازم ألاقي ليها (علشان تكرار)

⇐ للتأكيد إن حلنا صح هنجيب X(4)

$X(4) = e^{-J2\pi} + e^{-4J\pi}$

$= 1 + 1 = 2$ تكرار X(0)

$\Longrightarrow X(K) = \{2, -1-J, 0, -1+J\}$

- بعض الملاحظات العامة (المهمة)

- المفروض متقفلش القوس لأنه X(K) مكمله لـ ∞ لأنها periodic بتفضل تتكرر

- أول fundemental value لـ X(K) دايماً هيبقى ليها أعلى قيمة

- لازم كل قيمة Complex يكون ليها Conjugate

- اجيب قيم بعدد الـ n لأني هلاقيها بعد كده متكرره

$\boxed{2}$ plot The amplitude and phase spectrum?

$$X(K) = \{2, -1-J, 0, -1+J\}$$

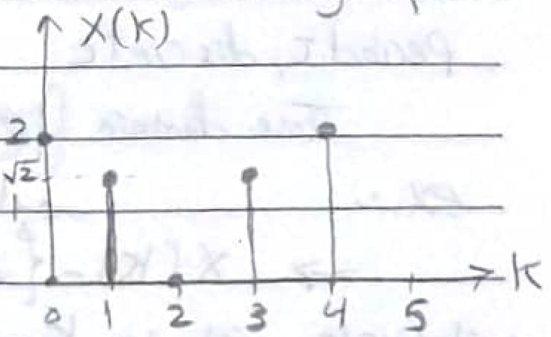
① The amplitude spectrum

$$|X(K)| = \sqrt{(Re)^2 + (Im)^2}$$

should be always positive

$$= \{\sqrt{(2)^2}, \sqrt{1+1}, 0, \sqrt{1+1}\}$$

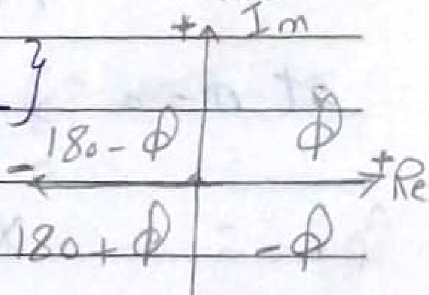
$$|X(K)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$$



Amplitude

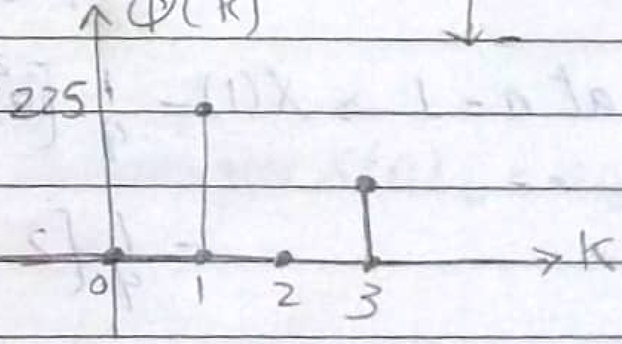
$\boxed{2}$ The phase spectrum $\phi(K)$

Real only	Imaginary only	Real ± Im (Complex)
$\phi = \begin{cases} 0 & +ve \\ 180 & -ve \end{cases}$	$\begin{cases} 90 & +ve \\ 270 & -ve \end{cases}$	$\phi = \tan^{-1}\frac{Im}{Re}$



$$\phi(K) = \{0, \tan^{-1}\frac{1}{1}, 0, \tan^{-1}\frac{1}{1}\}$$

$$\phi(K) = \{0, 225, 0, 135\}$$



Phase

$\boxed{3}$ Check The parseval's Theory?

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N}\sum_{n=0}^{N-1} |X(K)|^2$$

L.H.S $= 0 + 1 + 1 + 0 = 2$

R.H.S $= (4 + 2 + 0 + 2) * \frac{1}{4} = 2$

∴ Parseval's Theory is verified.

* To get inverse DFT (IDFT) of X(K):

$$X(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{J\frac{2\pi}{N}nk}$$

تحويل الnonperiodic من الFreq domain إلى periodic, discrete
في ال Time domain

ex :

هشتغل على نفس المثال السابق

$$\to X(K) = \{\overset{\uparrow}{2}, -1-J, 0, -1+J\}$$

هعتبرها non-periodic لأننا واخدين $K = 0 \to 3$ يعني واخدين cycle واحدة
يعني الsignal مش متكررة

$$\to X(n) = \frac{1}{4}\sum_{0}^{3} X(K\, e^{J\frac{2\pi}{4}nK}$$

$$X(n) = \frac{1}{4}\left[X(0)\, e^{J\frac{\pi}{2}*0} + X(1)\, e^{J\frac{\pi}{2}n} + X(2)\, e^{J\pi} + X(3)\, e^{J\frac{\pi}{2}3n}\right]$$

$$\text{at } n=0 \to X(0) = \frac{1}{4}\left[2*1 + (-1-J)*e^{J\frac{\pi}{2}*0} + 0 + (-1+J)\, e^{0}\right]$$

$$X(0) = \frac{1}{\cancel{4}}\left[\cancel{2} \cancel{-J} - 1 \cancel{-1} \cancel{+J}\right] = 0$$

$$\text{at } n=1 \to X(1) = \frac{1}{4}\left[\overset{2}{\cancel{0}}*1 + (-1-J)\, e^{J\frac{\pi}{2}} + 0 + (-1+J)\, e^{J\frac{3\pi}{2}}\right]$$

$$= \frac{1}{4}\left[2 + (-1-J)*J + 0 + (-1+J)*-J\right]$$

$$= \frac{1}{4}\left[2 \cancel{-J} + (\overset{-1}{\cancel{J^2}}) + 0 + (\cancel{J} - (\overset{-1}{J^2}))\right]$$

$$= \frac{1}{4}\left[2 + 0 + 2\right] = 1$$

$$X(1) = 1$$

CamScanner

at $n=2$:-

$$X(2) = \frac{1}{4}\left[2 + (-1-J)*e^{J\pi} + 0 + (-1+J)*e^{J3\pi}\right]$$

$$= \frac{1}{4}\left[2 + (-1-J)*-1 + 0 + (-1+J)*-1\right]$$

$$= \frac{1}{4}\left[2 + 1 + \cancel{J} + 0 + 1 - \cancel{J}\right] = 1$$

$$X(2) = 1$$

at $n=3$:-

$$X(3) = \frac{1}{4}\left[2 + (-1-J)e^{J*\frac{\pi}{2}*} + 0 + (-1+J)e^{J3\pi}\right]$$

$$= \frac{1}{4}\left[2 + (-1-J)(-J) + (-1+J)*-1\right]$$

$$= \frac{1}{4}\left[2 + J - J^2 + 1 - J\right] = \frac{1}{4}(2-2) = 0$$

$$X(3) = 0$$

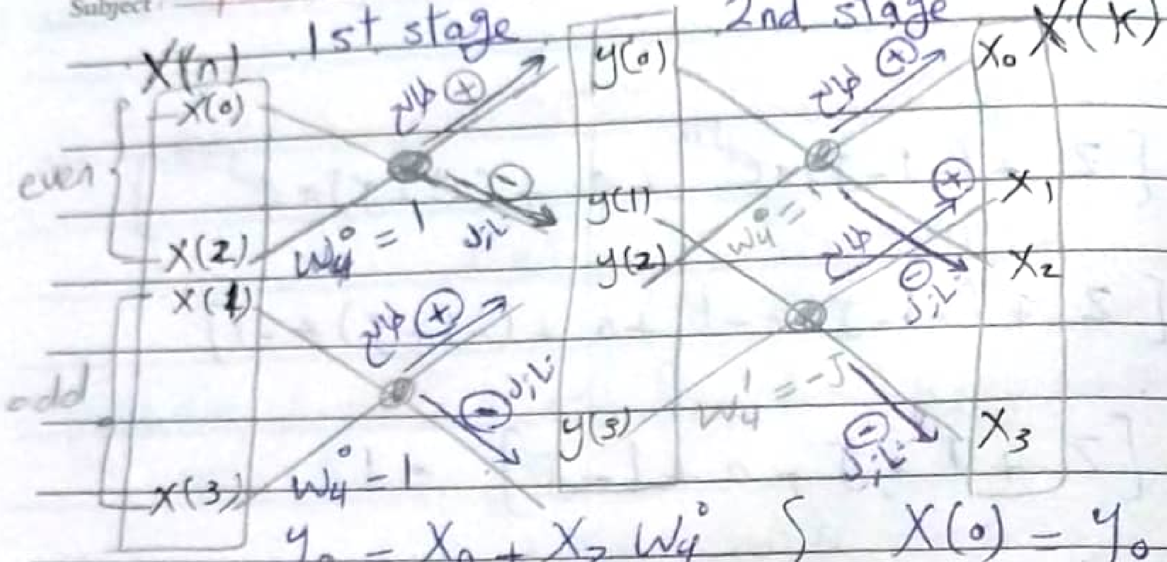
$$X(n) = \{0, 1, 1, 0\}$$

* مينفعش X(n) تطلع Complex

plot magnitude & phase for X(K)

Date : / /

Subject : Part(2) Chapter(2) FFT (Fast Fourier transform)



$y_0 = X_0 + X_2 W_4^0$
$y_1 = X_0 - X_2 W_4^0$
$y_2 = X_1 + X_3 W_4^0$
$y_3 = X_1 - X_3 W_4^0$

$X(0) = y_0 + y_2$
$X(1) = y_1 + (-J) y_3$
$X(2) = y_0 - y_2$
$X(3) = y_1 - (-J) y_3$

→ W_4^k → Twiddle factor , $W_N = e^{-J \frac{2\pi}{N} k}$, $W_4 = e^{-J\frac{\pi}{2}}$

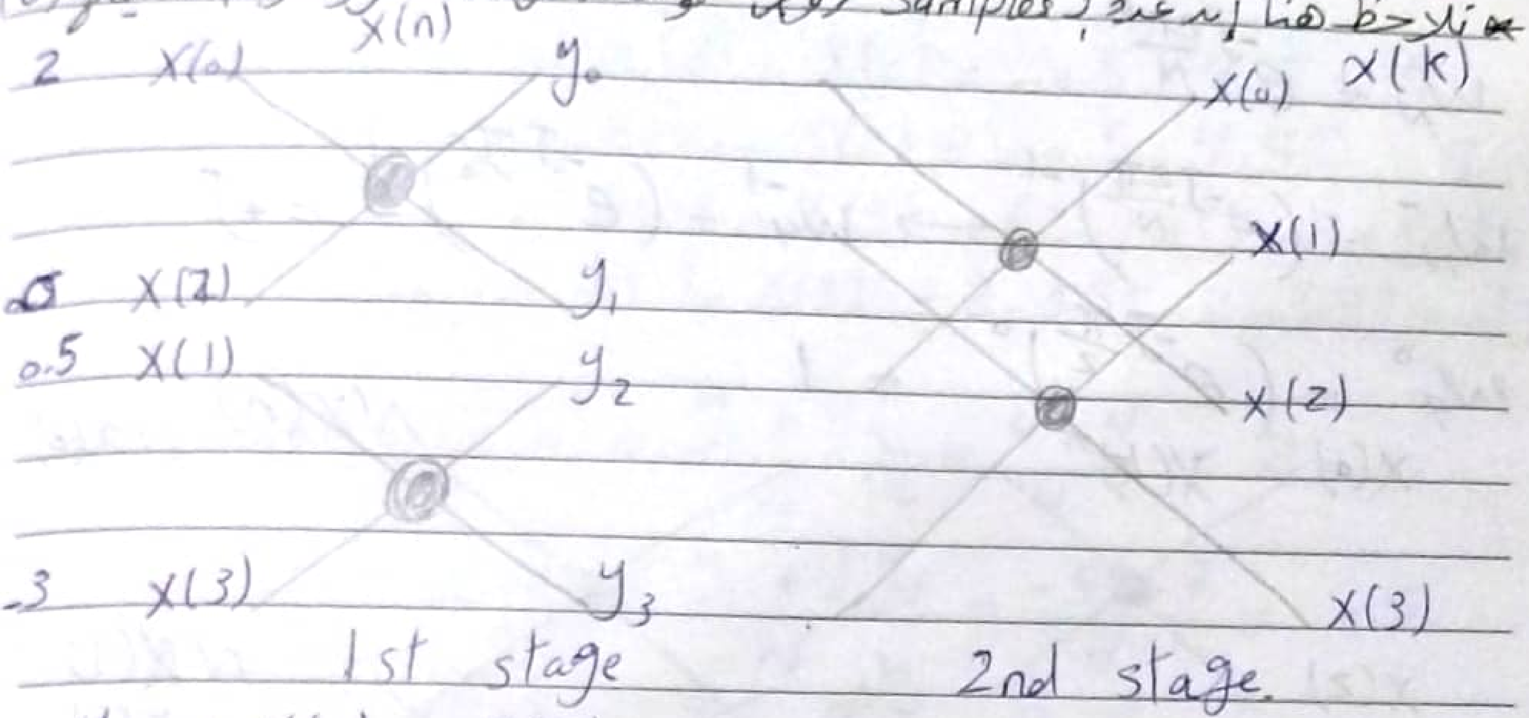
For $K = 0$: $W_4^0 = e^0 = 1$ $K = 1 \rightarrow W_4^1 = -J$ $(e^{-J\pi/2})$

ليه بتعمل ال FFT ؟ تسهيل خطوات الحل وفي الامتحان بيجبرني على الطريقة مش بيخيرني وممكن يطلب حل السؤال بالطريقتين

Called Method butterfly

Ex. Find $x(k)$ using FFT for: $x(n) = \{2, 0.5, 0, -3\}$

* نلاحظ هنا إن عدد ال samples زوجي لو كان فردي هنزود واحد فيبقى (0)



$y_0 = x(0) + x(2) = 2 + 0 = 2$

$y_1 = x(0) - x(2) = 2 - 0 = 2$

$y_2 = x(1) + x(3) = 0.5 + (-3) = -2.5$

$y_3 = x(1) - x(3) = 0.5 + 3 = 3.5$

$K=0 \rightarrow X(0) = y_0 + y_2 = 2 + (-2.5) = -0.5$

$K=1 \rightarrow X(1) = y_1 - (-J)\,y_3 = 2 + (-J * 3.5) = 2 - 3.5J$

$K=2 \rightarrow X(2) = y_0 - y_2 = 2 - (-2.5) = 4.5$

$K=3 \rightarrow X(3) = y_1 - (-J)\,y_3 = 2 - (-J * 3.5) = 2 + 3.5J$

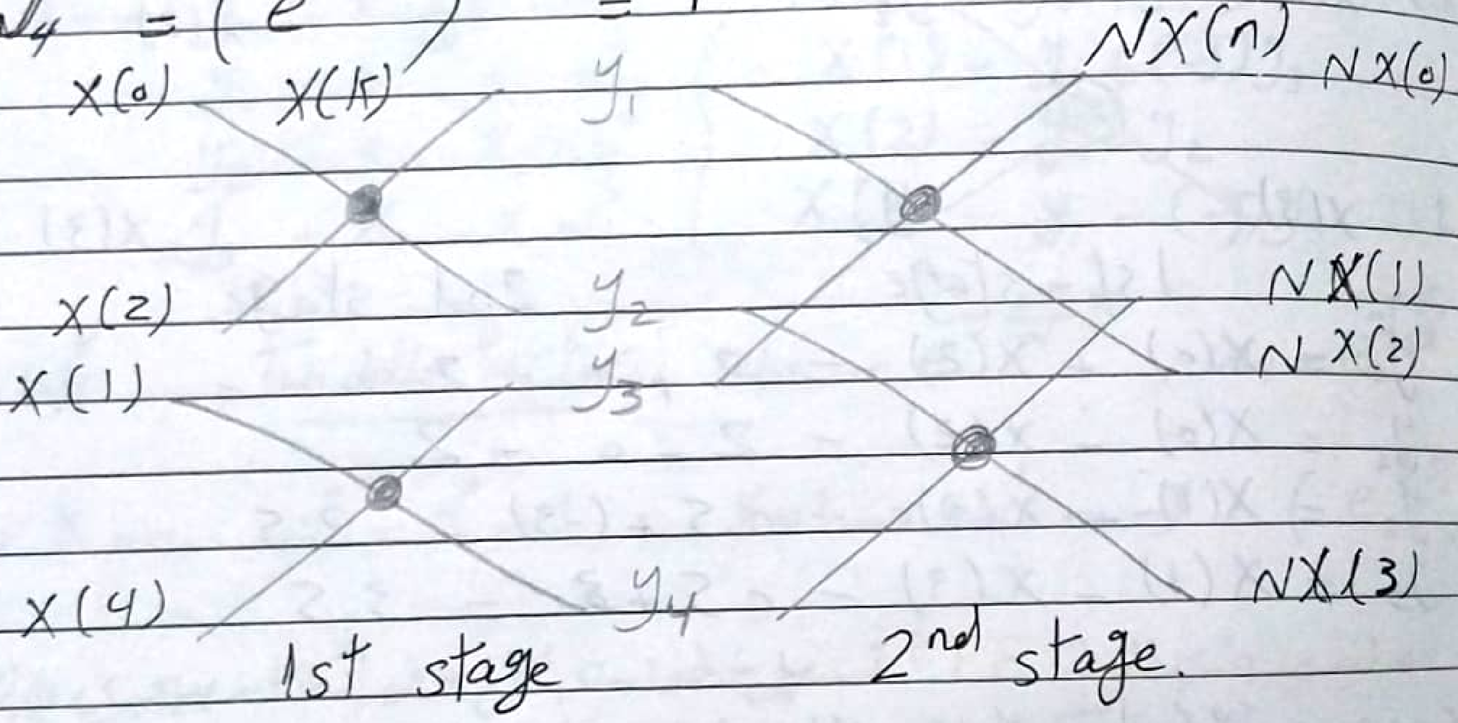
$\longrightarrow X(K) = \{-0.5,\ 2 - 3.5J,\ 4.5,\ 2 + 3.5J\}$

IFFT: $N x(n) = \sum_{k=0}^{N-1} X(k) W_N^{-nk}$

$W_N = e^{-J\frac{2nk}{N}}$

$W_N^{-1} = \left(e^{-J\frac{2\pi}{N}}\right)^{-1} \longrightarrow W_4^{-1} = \left(e^{-J\pi/2}\right)^{-1} = +J$

$W_4^{0} = \left(e^{-J\frac{\pi}{2}}\right)^{0} = 1$



$y_0 = X(0) + X(2)$
$y_1 = X(0) - X(2)$
$y_2 = X(1) + X(3)$
$y_3 = X(1) - X(3)$

$N x(0) = y_0 + y_2$
$N x(1) = y_1 + J y_3$
$N x(2) = y_0 - y_2$
$N x(3) = y_1 - J y_3$

Date: / / التاريخ: / /

Subject: موضوع الدرس:

→ for $X(K) = \{-0.5, 2-J3.5, 4.5, 2+J3.5\}$

Sol

1st stage:. $y_0 = X(0) + X(2) = -0.5 + 4.5 = 4$

$y_1 = X(0) - X(2) = -0.5 - 4.5 = -5$

$y_2 = X(1) + X(3) = (2-3.5J) + 2 + J\cancel{3.5} = 4$

$y_3 = X(1) - X(3) = 2 - 3.5J - 2 - 3.5J = -7J$

2nd stage:. $X(0) = \frac{y_0 + y_2}{N} = \frac{4+4}{4} = 2$

$X(2) = \frac{y_1 + J y_3}{N} = \frac{-5+7}{4} = 0.5$

$X(2) = \frac{y_0 - y_2}{N} = \frac{4-4}{4} = 0$

$X(3) = \frac{y_1 - J y_3}{N} = \frac{-5-7}{4} = -3$

$X(n) = \{2, 0.5, 0, -3\}$

→ check Parseval's Theory:.

$$\sum_{n=0}^{N-1} X^2(n) = \frac{1}{N} \sum_{K=0}^{N-1} |X(K)|^2$$

L.H.S → $4 + \frac{1}{4} + 0 + 9 = 13.25$

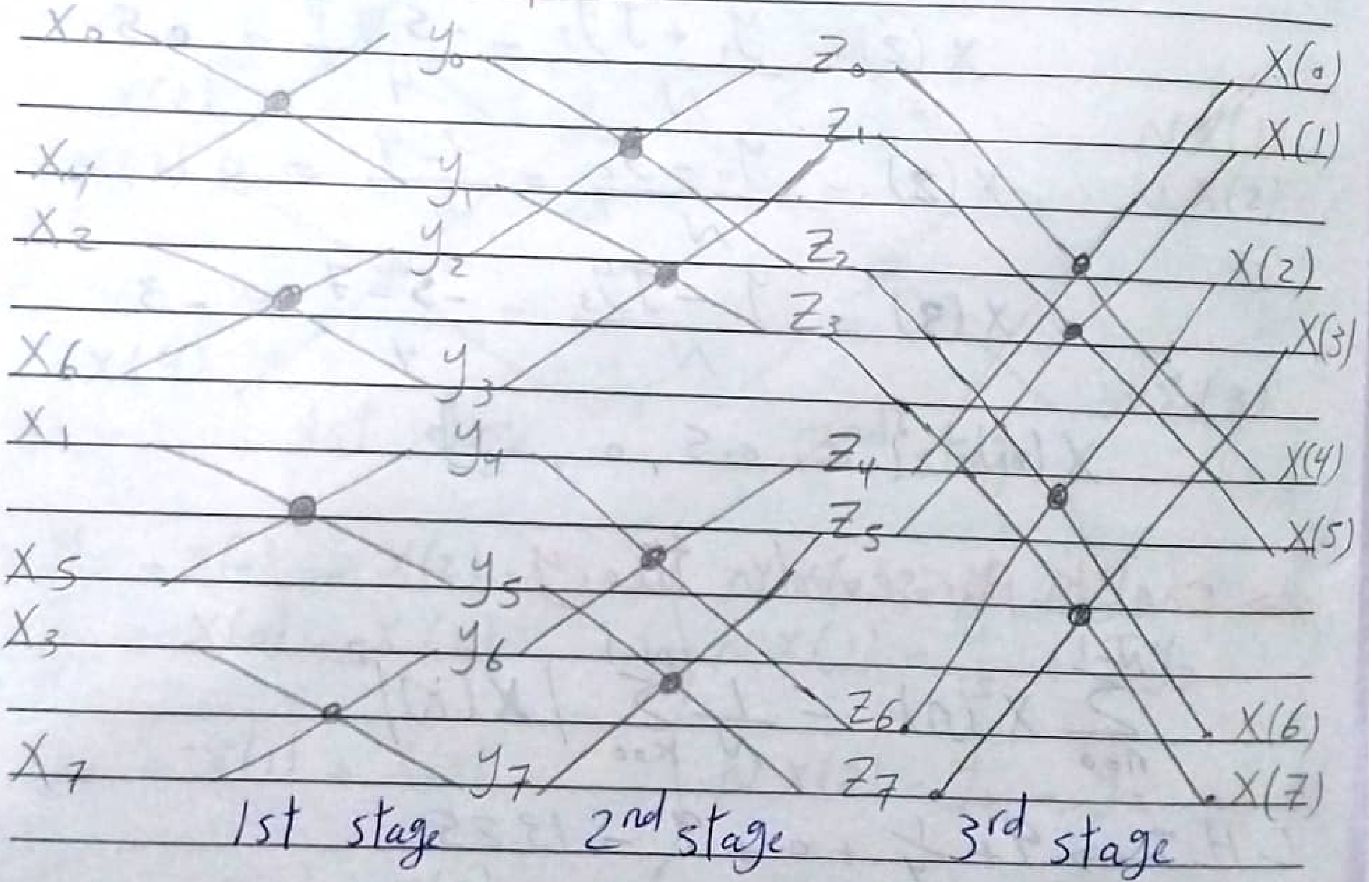
R.H.S → $\frac{1}{4}\left[\frac{1}{4} + \quad + \quad + \quad\right] = 13.25$

L.H.S = L.H.S تتحقق

→ Compare between FFT, DFT:-

	No of addition	No. of mult.
DFT	$N(N-1)$	N^2
FFT	$N\log_2 N$	$\frac{N}{2}\log_2 N$

FFT with 8 samples:-



$W_8^0 = 1$ $\quad$ $W_8^1 = \frac{1}{\sqrt{2}}$

$W_8^2 = -J$ $\quad$ $W_8^3 = \frac{-1}{\sqrt{2}}$

Date: 1 / 4 / 2023

Subject: Ch 3 Part (1)

التاريخ : / /

موضوع الدرس:

Correlation operation

used to Compare between Two signal to find The degree of symilarity.

← فكرة الرادار Auto Correlation

بحيث ان ال signal ترجع اتخبط في هدف ولما ترجع ال radar يقارنها بينها وبين ال signal اللي عنده وبكده احدد بعد الجسم وسرعته

← فكرة البصمة Cross Correlation

في البداية بتتعمل بيطلب منك أدخل أكثر من sample لايدك أو للوجه ولما تاتي تيجي تفتح الموبايل بياخد ال signal بتاعتك ويقارنها باللي عنده

Correlation types

اتنين مختلفين

Auto Correlation — يقارن ال signal بنفسها

Cross Correlation — يقارن 2signal مختلفين

$$r_{xx}(J) = \frac{1}{N_1}\sum_{n=0}^{N-1} X_1(n) . X_1(n+J)$$

$N_1 \rightarrow$ no of sample

$$\rho_{11}(J) = \frac{r_{11}(J)}{\frac{1}{N_1}\sum^{N-1} x_1^2(n)}$$

$\rho_{11}(J) \rightarrow 1$ متشابهين

$\rho_{11}(J) \rightarrow -1$ عكس في ال phase

$$r_{12}(J) = \frac{1}{N_{new}}\sum_{n=0}^{N-1} x_1(n) . x_2(n+J)$$

or $r_{21}(J)$

Two types :.

1. linear Cross Corre

$N_{new} = N_1 + N_2 - 1$

2. Cyclar cross

$N_{new} = N_{large}$

$$\rho_{12}(J) = \frac{r_{12}(J)}{\frac{1}{N_{new}}\sqrt{\sum x_1^2(n) . x_2^2}}$$

→ Auto Correlation :-

Ex :. $x_1(n) = \{2, 3, 0, -1, 3\}$, $x_2(n) = \{1, 4, 0, 2\}$

Req :.

1. Find The auto-Correlation of $x_1(n)$, $x_2(n)$
2. Find the Cross Correlation
3. Draw The relation between $r_{12}(J)$, J

Date: / /

Subject:

Solution: $r_{11}(J) = \frac{1}{N_1} \sum_{n=0}^{N-1} X_1(n) \cdot X_1(n+J)$

$N_1 = 5$

$X_1(n)$	2	3	0	-1	3	J	$r_{11}(J)$	$\rho_{11}(J)$
	2	3	0	-1	3	0	$\frac{23}{5} = 4.6$	4.6/4.6
	3	0	-1	3	0	1	0.6	0.6/4.6
$X_1(n+J)$	0	-1	3	0	0	2	-0.6	-0.6/4.6
	-1	3	0	0	0	3	1.4	1.4/4.6
	3	0	0	0	0	4	1.2	1.2/4.6

$$\rho_{11}(J) = \frac{r_{11}(J)}{\frac{1}{N_1} \sum_{n=0}^{N-1} X_1^2(n)}$$

$$\frac{1}{N_1} \sum_{n=0}^{N-1} X_1^2(n) = \frac{1}{5}\left(2^2 + 3^2 + (-1)^2 + (3)^2\right) = \frac{23}{5}$$

ودة طبيعي لأنه هيكون symetric فأول نشتغل بس عند J=0

$\boxed{2}$ Cross Correlation:

~~$\boxed{1}$ linear:~~ $r_{12}(J) = \frac{1}{N_{new}} \sum_{n=0}^{N-1} X_1(n) \cdot X_2(n+J$

$\boxed{1}$ linear $N_{new} = N_1 + N_2 - 1 = 8$ sample

$$\rho_{12} = \frac{r_{12}(J)}{\frac{1}{N_{new}} \sqrt{\sum_{n=0}^{N-1} X_1^2(n) \cdot \sum_{n=0}^{N-1} X_2^2(n)}}$$

Date : / /

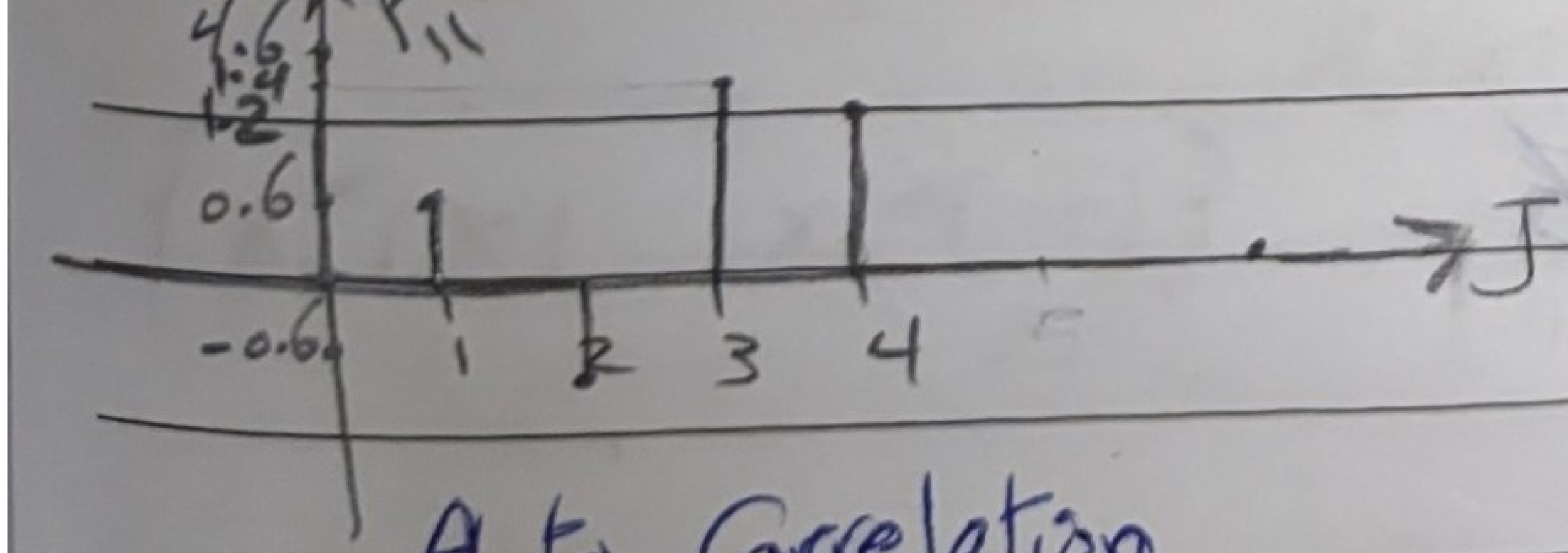
Subject :

$X_1(n)$	2	3	0	-1	3	0	0	0	J	$r_{12}(J)$	$\rho_{12}(J)$
	1	4	0	2	0	0	0	0	0	1.5	1.5/2.7
	4	0	2	0	0	0	0	1	1	1	1/2.7
	0	2	0	0	0	0	1	4	2	0.75	0.75/2.7
$X_2(n+J)$	2	0	0	0	0	1	4	0	3	0.5	0.5/2.7
	0	0	0	0	1	4	0	2	4	0.375	0.375/2.7
	0	0	0	1	4	0	2	0	5	1.375	1.375/2.7
	0	0	1	4	0	2	0	0	6	-0.5	-0.5/2.7
	0	1	4	0	2	0	0	0	7	1.125	1.125/2.7

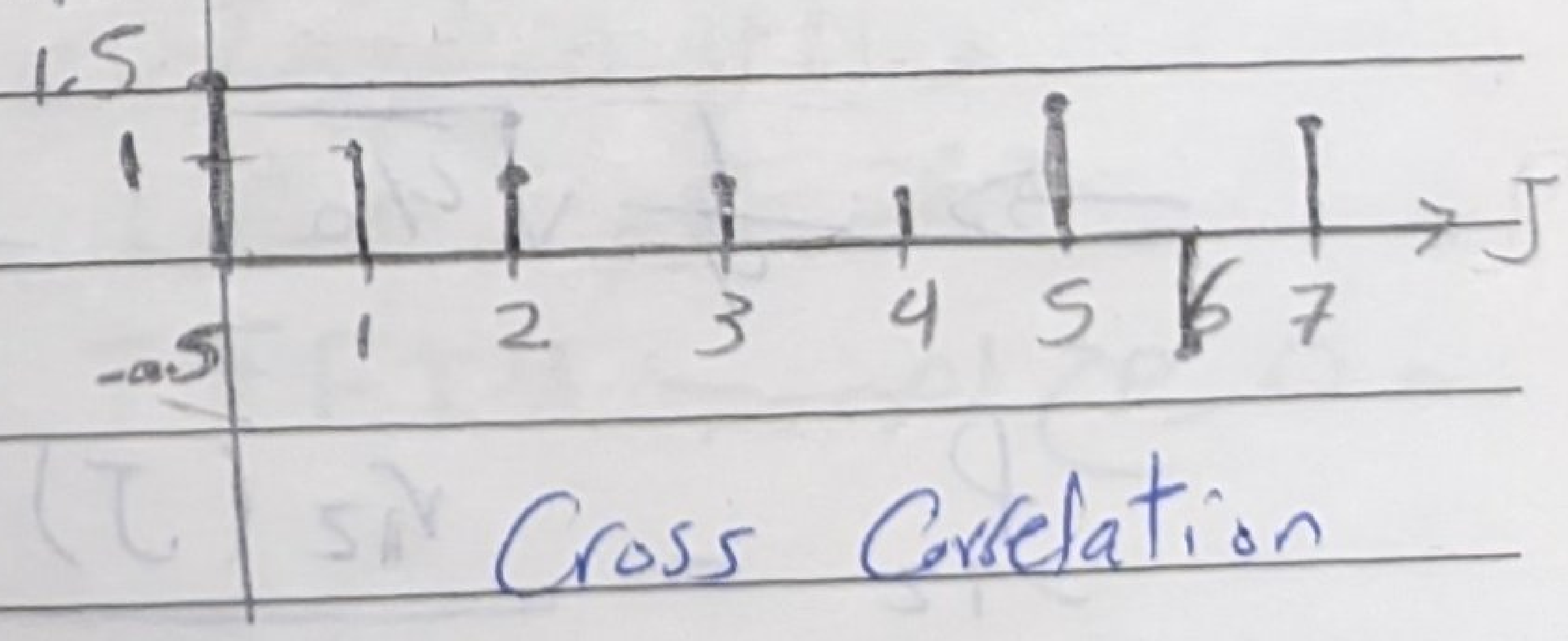
$$\rho_{12} \quad \frac{r_{12}(J)}{\frac{1}{8}\sqrt{(2^2+3^2+(-1)^2+(3)^2][(1)^2+(4)^2+(2)^2 \ldots}}$$

$$\rho_{12} = \frac{r_{12}(J)}{2.74}$$

$\boxed{3}$ Relation between $r_{12}(J)$ and J :-



Auto Correlation



Cross Correlation

- Correlation Theory :-

$$r_{12}(J) = \frac{1}{N_{new}} F^{-1}\left[X_1^*(k) \cdot X_2^*(k)\right]$$

$X_1^*(k) \longrightarrow$ FFT $X_1(n)$, $X_2^*(k) \longrightarrow$ FFT $X_2(n)$

$F^{-1} \longrightarrow$ IFFT of $\left[X_1^*(k) \cdot X_2^*(k)\right]$

Date : / /

Subject :

Example: $X_1(n) = \{2, 0, -1\}$, $X_2(n) = \{3, 1\}$

Find $r_{12}(J)$ by ① Direct Method.
② Correlation Theory.

$N_1 = 3$ Sample , $N_2 = 2$ sample.

① using linear Cross Correlation.

$N_{new} = 3 + 2 - 1 = 4$ Sample.

	$X_1(n)$				J	r_{12}	ρ_{12}
	2	0	-1	0			
$X_2(n+J)$	3	1	0	0	0	1.5	$\simeq 1$
	1	0	0	3	1	0.5	0.5/1.58
	0	0	3	1	2	-0.75	-0.75/1.58
	0	3	1	0	3	-0.25	-0.25/1.76
	3	1	0	0			

← sequence repeated

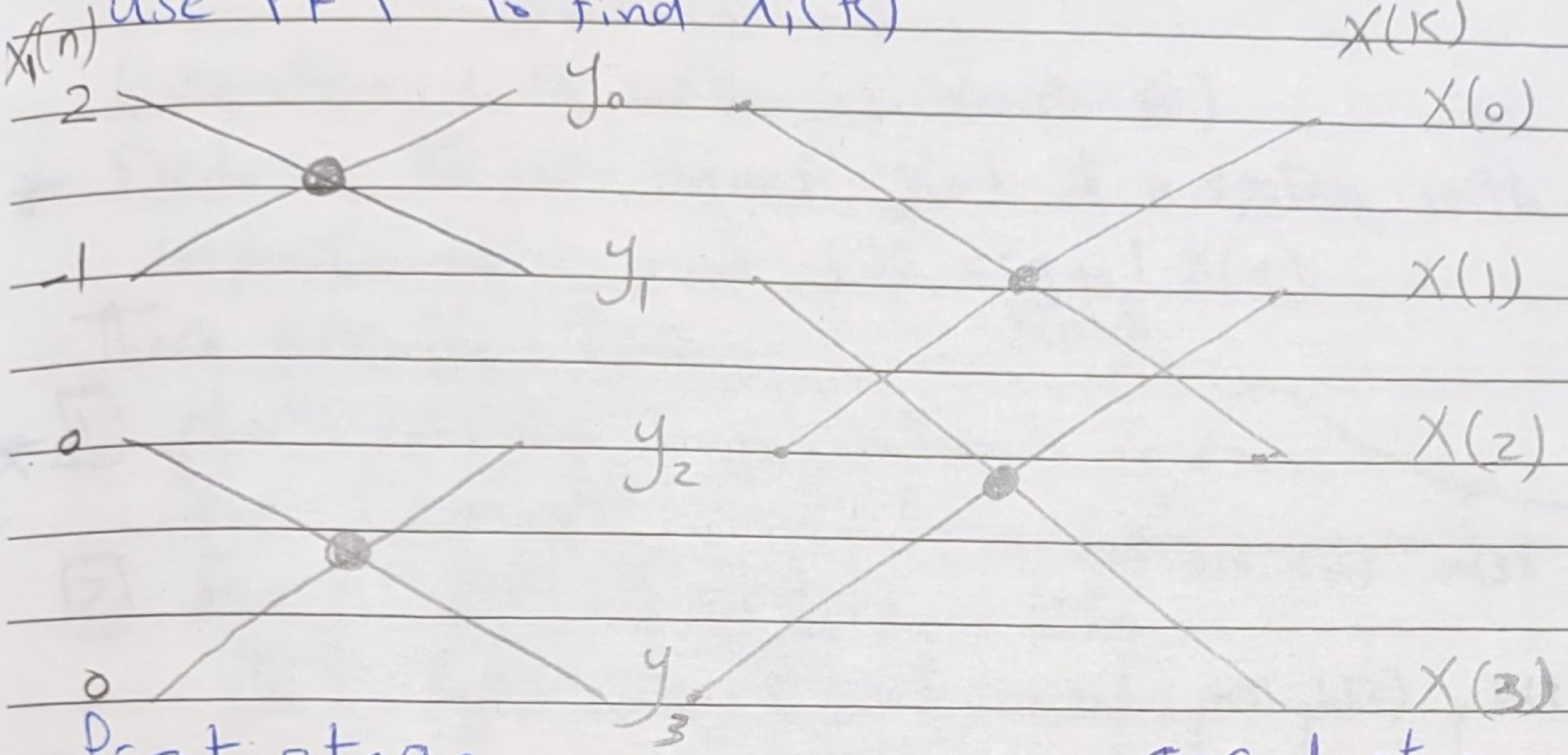
$$\rho_{12} = \frac{r_{12}(J)}{\frac{1}{4}\sqrt{[2^2 + (-1)^2] \cdot [3^2 + 1^2]}}$$

$$\Rightarrow \frac{1}{4}\sqrt{40} = 1.58$$

$$\rho_{12} = \frac{r_{12}(J)}{1.58}$$

② Correlation Theory:

use FFT to find $X_1(K)$



first stage:

$y_0 = -1 + 2 = 1$

$y_1 = -1 - 2 = -3$

$y_2 = 0 + 0 = 0$

$y_3 = 0 - 0 = 0$

second stage:

$X(0) = y_0 + y_2 = 1 + 0 = 1$

$X(1) = -3 - Jy_3 = -3 + 0$

$X(2) = 1 - 0 = 1$

$X(3) = -3 + Jy_3 = -3$

Then get Conjecate $X(K) = \{1, 0, 1, -3\}$

$X_1^*(K) = \{1, 0, 1, -3\}$ بيغير اشاره لـ J

وبالمثل هنعمل كونجم حق $X_2(n)$

هنضربهم في بعض

$\{X_1^*(K) \cdot X_2^*(K)\} \rightarrow$ هنرجع اصلها بالـ IFFT sec No (6)

Date : / /　　　　　　　　　　　　　　　　　　　التاريخ : / /

Subject : Ch(3) part (2)　　Convolution　　موضوع الدرس:

It is how the input interact with the system to produce o/p

(system with unknown response)

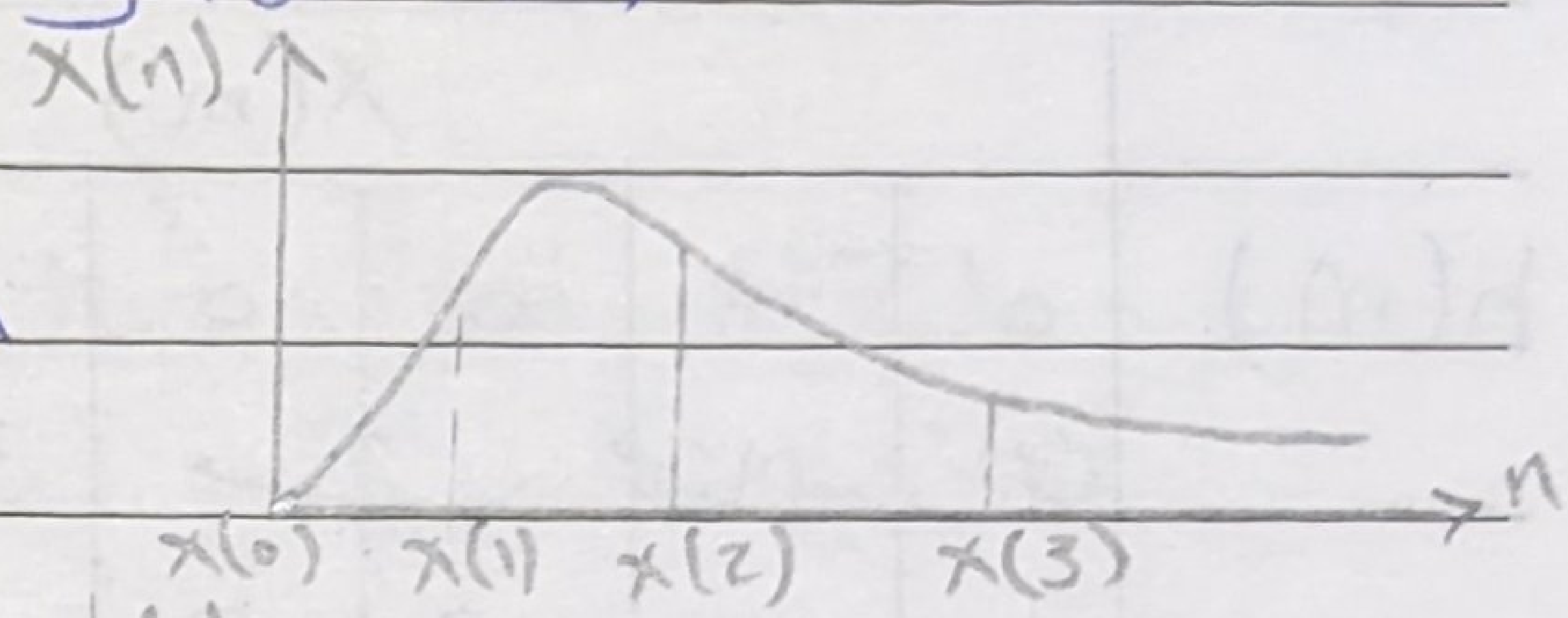
* Deduce the o/p signal y(n) of a system with responce h(n) and I/p signal x(n)

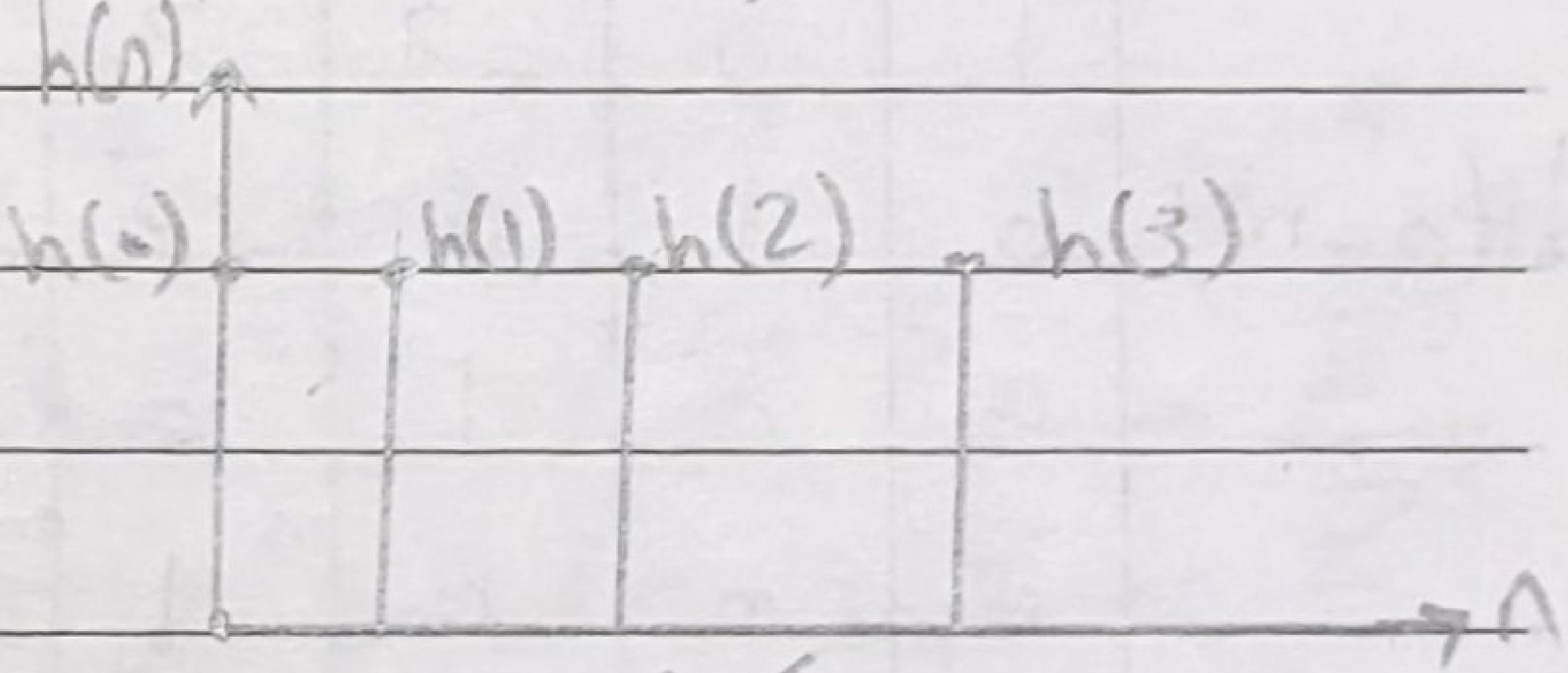
There are Two Types :-

* [1] Mathematical Convolution for Discrete Signal

[2] Analytical Convolution for Continous Signal

طلعى الترم ده



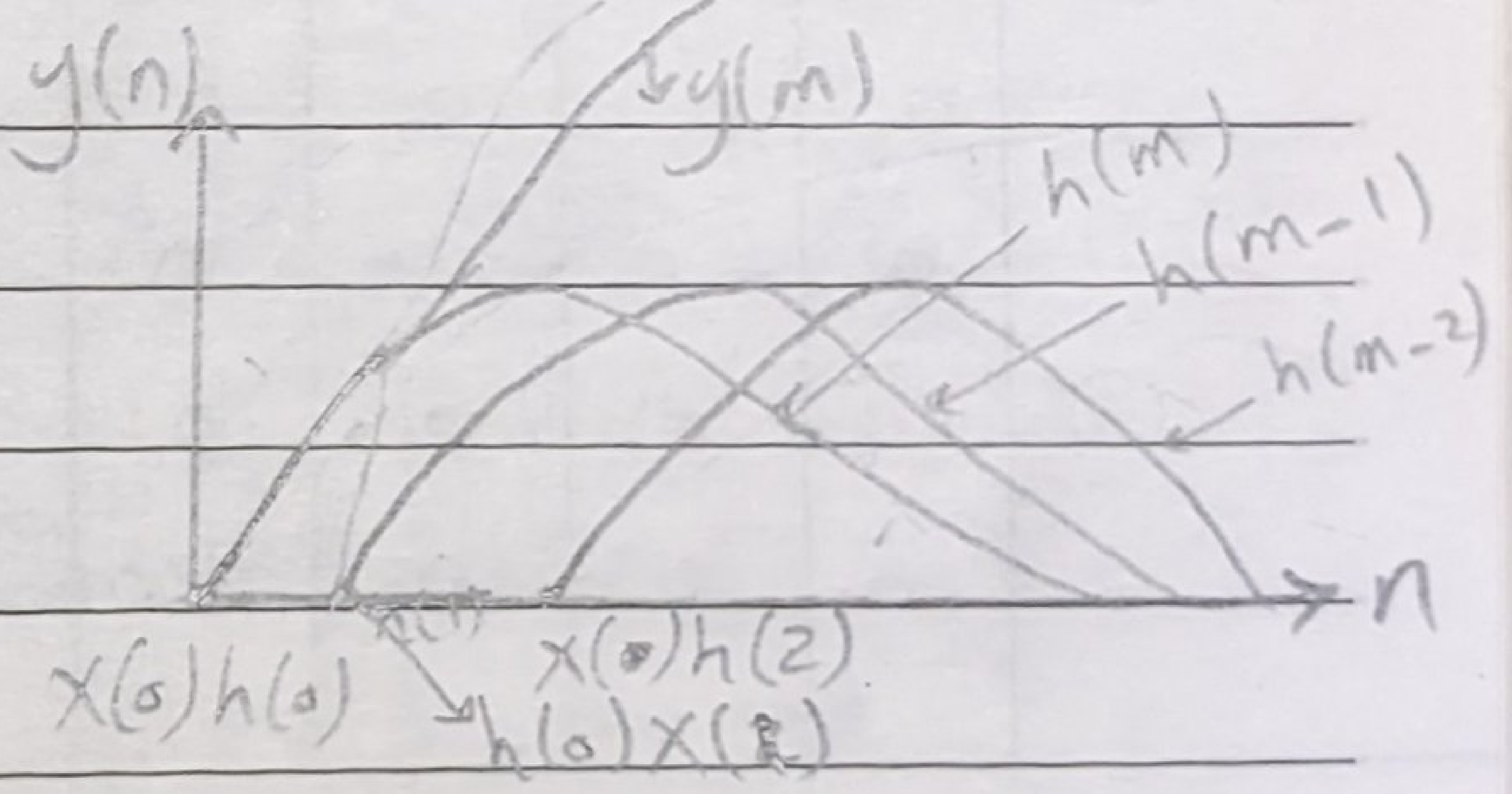


→ for mathematical Convolution

$$y(n) = \sum_{m=0}^{n} h(m)\,x(n-m)$$

$$\text{oR} = \sum_{m=0}^{n} x(m)\,h(n-m)$$

كأنها عليه حرب



$y(0) = x(0)\,h(0)$

$y(1) = x(1)\,h(0) + x(0)\,h(1)$

$y(2) = x(2)\,h(0) + x(1)\,h(1) + x(0)\,h(2)$

⋮

$y(n) = x(n)\,h(0) + x(n-1)\,h(1)$ --

$\Rightarrow y(n) = \sum h(m).\,x(n-m)$

oR $y(n) = \sum x(m)\,h(n-m)$

$$\# \quad y(n) = \sum_{m=0}^{n} x(m)\,h(n-m)$$

Scanned by TapScanner

Date : / / التاريخ : / /

Subject : موضوع الدرس:

ex :. Given $x(m) = \{2, 1, 0, -1\}$

$h(m) = \{3, 2, -2, 0, 1\}$ find o/p

يا ما تثبت ال X وتعكس لـ h يا ما العكس

$N_{new} = 4 + 5 - 1 = 8$ sample.

	x(m)								n	y(n)
h(m)	0	0	0	0	2	1	0	-1		
	ثابت كما هو				3	2	-2	0	1	
	1	0	-2	2	3	0	0	0	0	6
h(0-m)	0	1	0	-2	2	3	0	0	1	7
	0	0	1	0	-2	2	3	0	2	-2
	0	0	0	1	0	-2	2	3	3	-5
	0	0	0	0	1	0	-2	2	4	0
	0	0	0	0	0	1	0	-2	5	3
	0	0	0	0	0	0	1	0	6	0
	0	0	0	0	0	0	0	1	7	-1

Date 2/5/, 3/5 التاريخ : / /

Subject : Chap. (4) Z Transform موضوع الدرس:

هو Domain بحول ليه الـ signal لتسهيل العمليات زي لابلاس
مثلا الـ integration في Time domain معقد فبحول لـ laplace domain
الـ laplace بيتعامل مع الـ Continous أما الـ Z مع الـ Discrete

→ to Transform to Z-Transform:

$$X(Z) = \sum_{n=-\infty}^{\infty} x(n) Z^{-n}$$ n = No. of sample

Ex.: $x(n) = \{3, 1, 0, -2, 3, 4\}$ find $X(Z)$

① $X(Z) = \sum_{n=0}^{5} x(n) Z^{-n}$

$= 3Z^{0} + 1Z^{-1} + 0Z^{-2} + (-2)Z^{-3} + 3Z^{-4} + 4Z^{-5}$

$= 3 + Z^{-1} - 2Z^{-3} + 3Z^{-4} + 4Z^{-5}$ $\ast \frac{Z^{5}}{Z^{5}}$

← علشان تخلي شكلها أفضل من غير السالب

$\longrightarrow X(Z) = \frac{1}{Z^{5}}\left[3Z^{5} + Z^{4} - 2Z^{2} + 3Z^{1} + 4\right]$

② find ROC: (Region of Convergence)

All values of Z which make $X(Z) \neq \infty$

$\boxed{X(Z) = \frac{1}{0} = \infty}$ القيم اللي بوصلها اللي آخلي $X(Z)=\infty$ هي صفر ⟸

$\Longrightarrow$ ROC All values except at $Z = 0$

③ poles and Zeros:

poles ← أصفار المقام (القيم اللي تخلي المقام = صفر)
Zeros ← أصفار البسط (〃 〃 〃 البسط = 〃)

Ex: $x(n) = \{1, -2, -24\}$ plot Poles & Zeros

$$X(Z) = \sum_{n=0}^{2} x(n) Z^{-n}$$

$$= 1 \times Z^{0} + (-2 Z^{-1}) + (-24 Z^{-2})$$

$$= 1 - 2Z^{-1} - 24Z^{-2} \quad * \frac{Z^2}{Z^2}$$

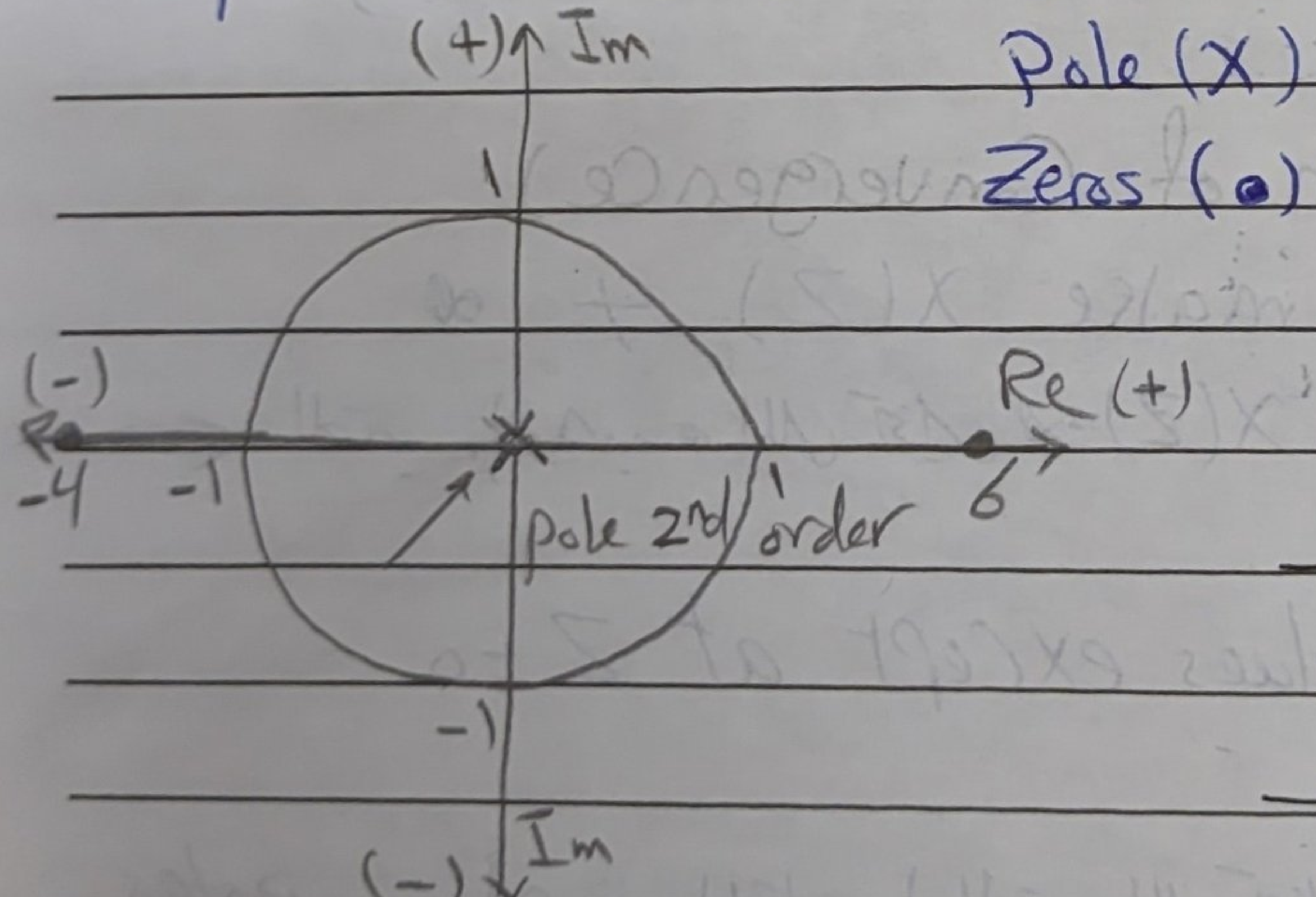
$$\rightarrow X(Z) = \frac{Z^2 - 2Z - 24}{Z^2}$$

To find poles & Zeros:

$$\rightarrow X(Z) = \frac{(Z+4)(Z-6)}{Z^2}$$

Zeros at $Z = -4$, $Z - 6$

poles at $Z_1 = 0$, $Z_2 = 0$ Two poles 2nd order.



→ All poles inside unit Circle = stable system.

→ one Pole at least on Circle = Critical stable.

→ one pole at least outside Circle → unstable.

in This system is stable.
All pole inside circle.

EX :
$$X(Z) = \frac{1-3Z^{-1}-18Z^{-2}}{1-Z^{-2}}$$

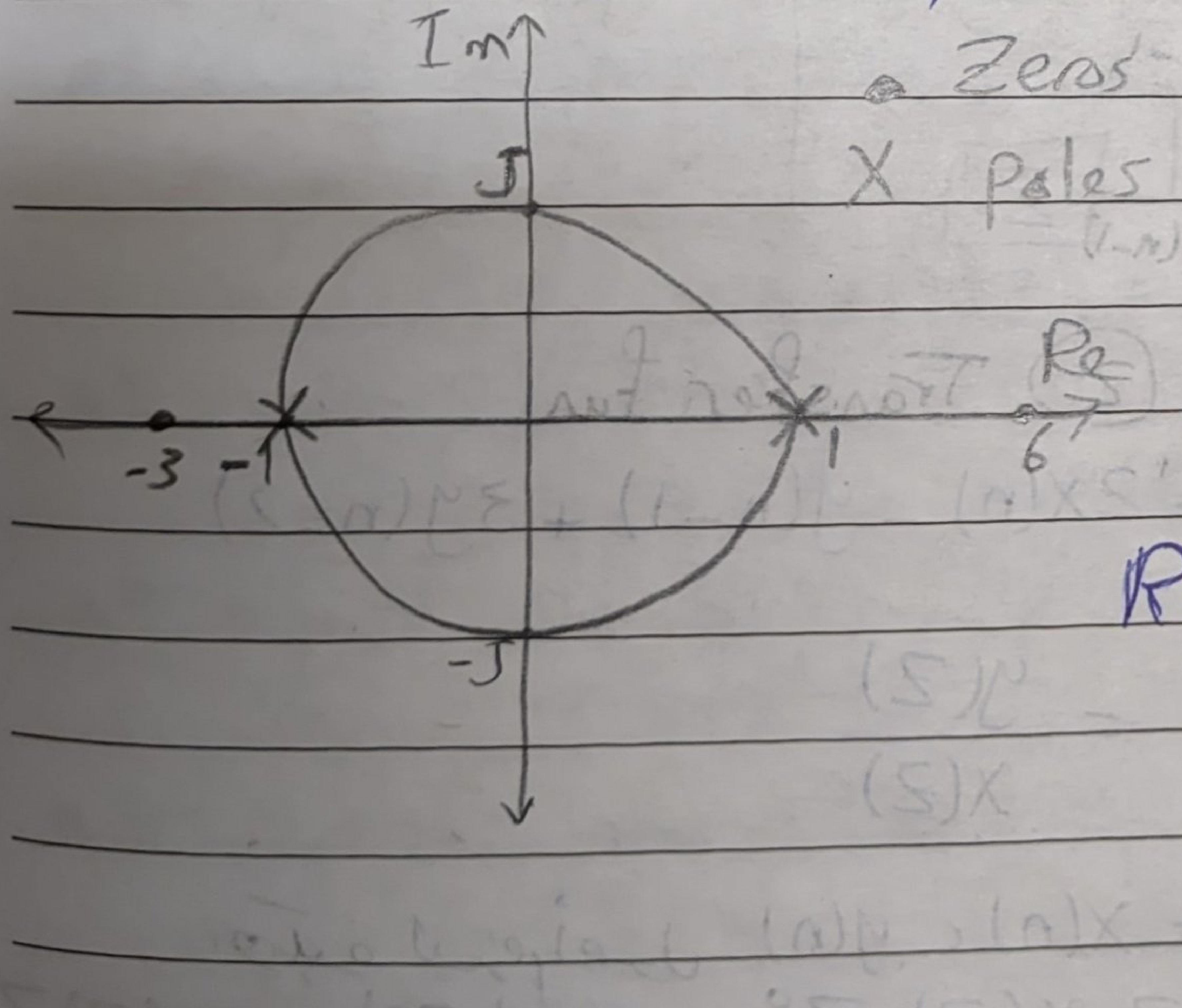
Find poles and Zeros Then plot Them

$$X(Z) = \frac{1-3Z^{-1}-18Z^{-2}}{1-Z^{-2}} * \frac{Z^2}{Z^2}$$

$$X(Z) = \frac{Z^2-3Z-18}{Z^2-1}$$

$$X(Z) = \frac{(Z-6)(Z+3)}{(Z-1)(Z+1)}$$

poles at $Z=1$, $Z=-1$
Zeros at $Z=6$, $Z=-3$



System is Critical stable.

ROC : All value except at $Z = \{1, -1\}$

Date : / /

Subject :

التاريخ : / /

موضوع الدرس:

shifting property (Delay property):-

$$x(n) \rightarrow \boxed{Z^{-1}} \rightarrow x(n-1)$$

$$X(Z)\,Z^{-k} \rightleftarrows x(n-k)$$

وبعده لو ضربنا $X(n).Z^{1}$ ← هيعمل Delay = 1

$X(n).Z^{k}$ ← هيعمل delay = k

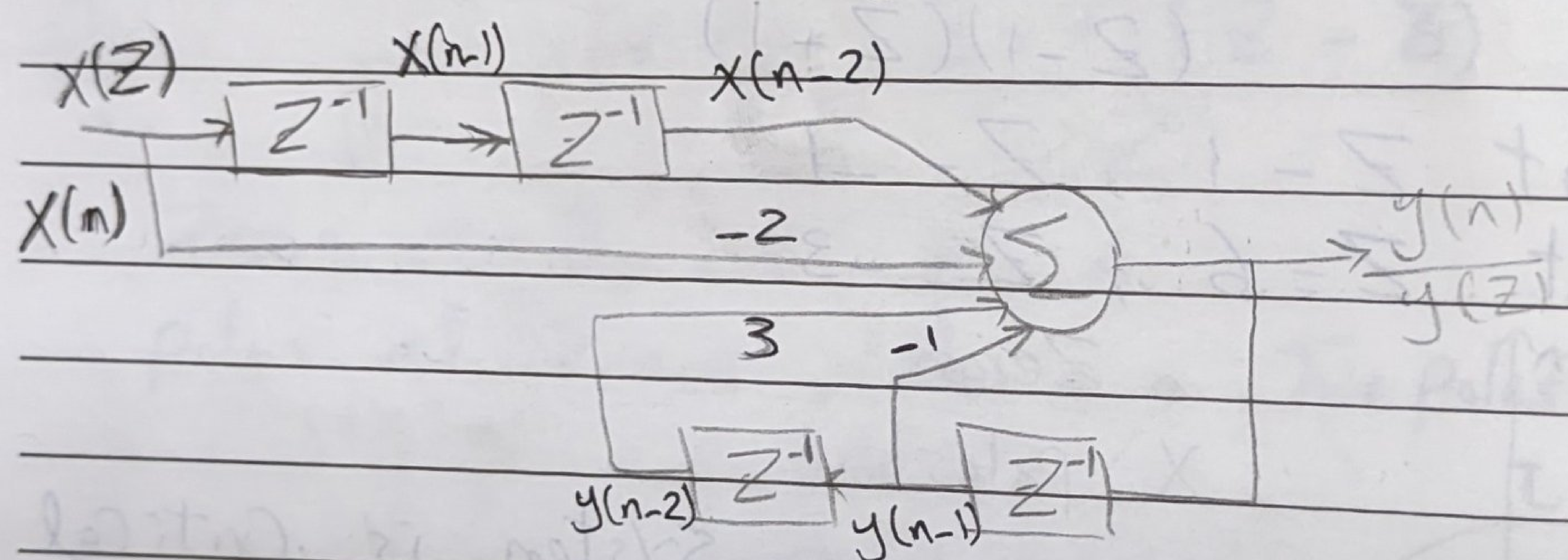
→ Difference equation:

علاقه بين I/P ، o/p بس في ال Time domain

→ Transfer function:

علاقه بين I/P ، o/p بس في ال Z domain

Ex → from The block diagram.



Find ① Difference eq , ② Transfer fun.

$\boxed{1}$ $y(n) = x(n-2) - 2x(n) - y(n-1) + 3y(n-2)$

$\boxed{2}$ $y(Z) \Rightarrow T.f = \frac{y(Z)}{X(Z)}$

$X(Z)Z^{-k} \rightarrow x(n-k) \Leftarrow x(n) ، y(n)$ وبعده نترجم أصول

$\rightarrow y(Z) = X(Z)Z^{-2} - 2X(Z)Z^{0} - y(Z)Z^{-1} + 3y(Z)Z^{-2}$

$= X(Z)Z^{-2} - 2X(Z) - y(Z)Z^{-1} + 3y(Z)Z^{-2}$

$\rightarrow T.f = \frac{y(Z)}{X(Z)} = \frac{-2 + Z^{-2}}{1 + Z^{-1} - 3Z^{-2}}$

Date : / /

التاريخ : / /

Subject :

موضوع الدرس:

← عندي ٣ أنواع من ال signal هنتعامل معاهم من ناحية ال responce وازاي بتعامل معاهم

① Impulse ② unit step ③ Ramp

* when I/P is Impulse with $x(n) \xrightarrow{\text{تتحول}} \delta(n)$

$y(n) \longrightarrow h(n)$

from differerence eq at previous example

$$y(n) = x(n-2) - 2x(n) - y(n-1) + 3y(n-2)$$

$$\longrightarrow h(n) = \delta(n-2) - 2\delta(n) - h(n-1) + 3h(n-2)$$

$$\text{as } \delta(n) = \begin{cases} 0 & \text{at } n \neq 0 \\ 1 & \text{at } n = 0 \end{cases}$$

→ Calculate h(n) for 5 Sample — طالما معنديش intial condition

at $n=0 \rightarrow h(0) = \cancel{\delta(0-2)} - 2\delta(0) - \cancel{h(0-1)} + 3\cancel{h(0-2)}$

= صفر لانها مش موجوده عند -2

$\longrightarrow h(0) = -2 * 1 = -2$

at $n=1 \rightarrow h(1) = \overset{0}{\delta(1-2)} - 2\overset{0}{\delta(1)} - \overset{h(0)}{h(1-1)} + 3\overset{0}{h(1-2)}$

$\rightarrow h(1) = -2 * 0 + 2 = 2$

at $n=2 \rightarrow h(2) = \delta(2-2) - 2\delta(2) - \overset{h(1)}{h(2-1)} + 3\overset{h(0)}{h(2-2)}$

$h(2) = 1 - 2 + 3 * -2 = -7$

at $n=3 \rightarrow h(3) = \delta(1) - 2\delta(3) - h(2) + 3h(1)$

$h(3) = -(-7) + 3*2 = 13$

$$h(n) = \{-2, 2, -7, 13\}$$

Date : / /	التاريخ : / /

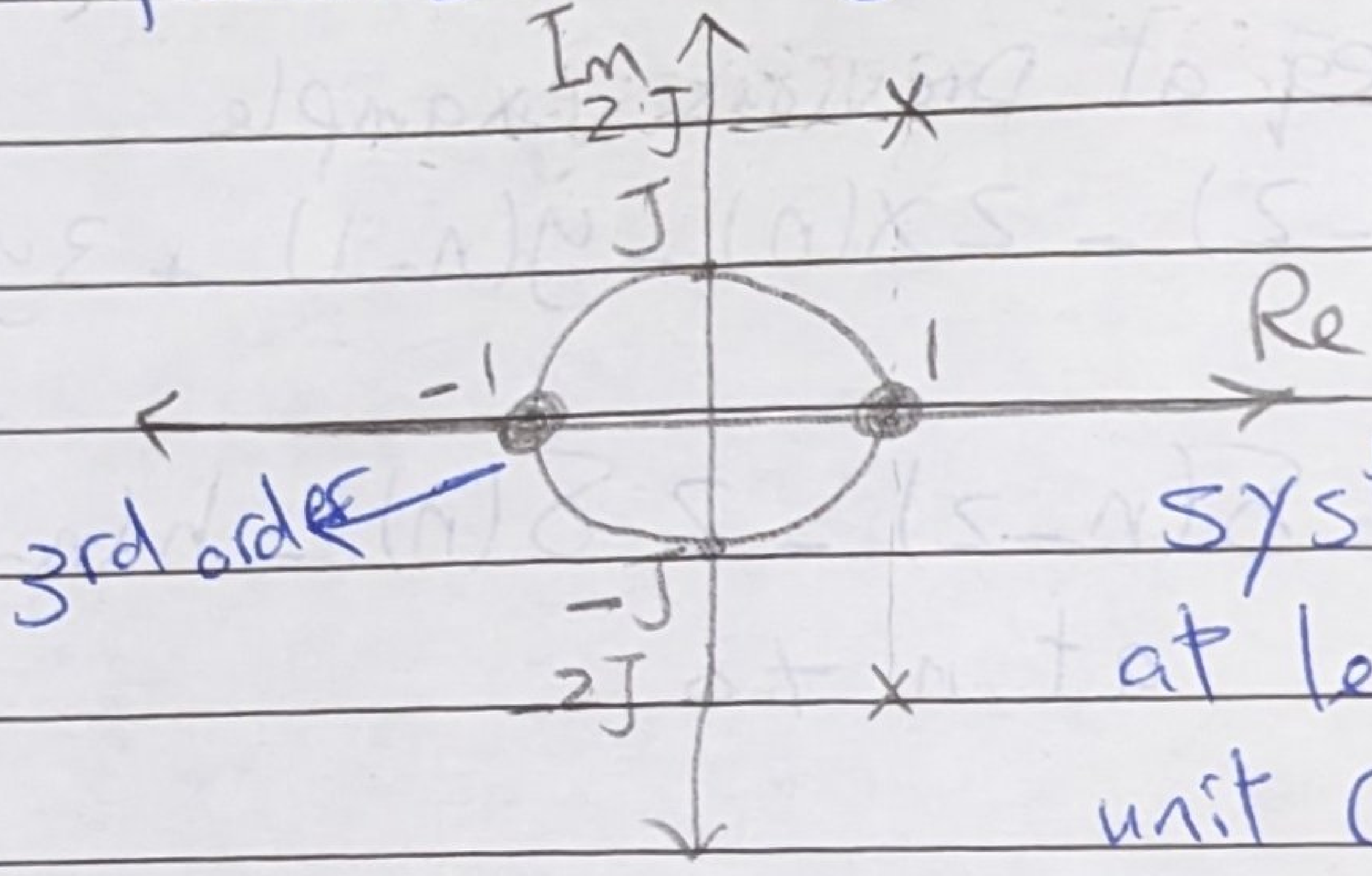
Subject :	موضوع الدرس:

EX: A system has a transfer fun with pole at $Z = \{0.5, 1 \pm 2J\}$, Zeros at $\{-1, 1\}$ 3rd order ↓

لاحظ عشان ال Complex pole يبقى فيه Conjecate تباعه

لاحظ مش هيجيبوا في مسألة فيها تحليل وفيها Complex

$\boxed{1}$ plot pole - Zero diagram:



system unstable at least one pole outside unit circle (Two pole outside)

* علشان اتغلب على ال unstability ← لو فيه قوس $\frac{(Z+3)}{(Z+3)}$ بسط ومقام زي بعض هيعملو Cancellation لبعض ويكمل ال system يصبح stable والطريقة دي تسمى ال pole Zero Cancellation

$\boxed{3}$ Drive T.f:

$$H(Z) = \frac{(Z-(-1))^3 (Z-1)}{(Z-0.5)(Z-(1-2J))(Z+(1+2J))}$$

$$H(Z) = \frac{(Z+1)^3 (Z-1)}{(Z-0.5)(Z-1+2J)(Z-1-2J)}$$

$\boxed{4}$ Difference equation:

↳ $\boxed{y(n)}$

$\overset{5}{(Z-(1+2J))(Z-(1-2J))}$ $Re^2 + Im^2$

$Z^2 - Z - \cancel{2JZ} - Z + \cancel{2JZ} + 5$

$Z^2 - 2Z + 5$

$(Z^2+2Z+1)\ (Z^2-1)$

$$H(Z) = \frac{(Z+1)^2 (Z+1)(Z-1)}{(Z-0.5)(Z^2-2Z+5)}$$

$$H(Z) = \frac{Z^4 + 2Z^3 - 2Z - 1}{Z^3 - 2.5Z^2 + 6Z - 2.5} * \frac{Z^{-4}}{Z^{-4}}$$

Scanned by TapScanner

Date : / /

التاريخ : / /

Subject :

موضوع الدرس:

$$\frac{y(Z)}{X(Z)} = \frac{1 + 2Z^{-1} - 2Z^{-3} - Z^{-4}}{Z^{-1} - 2.5Z^{2} + 6Z^{-3} - 2.5Z^{-4}}$$

$$y(Z)Z^{-1} - 2.5Z^{-2}y(Z) + 6y(Z)Z^{-3} - 2.5y(Z)Z^{-4}$$
$$= X(Z) + 2X(Z)Z^{-1} - 2X(Z)Z^{-3} - X(Z)Z^{-4}$$

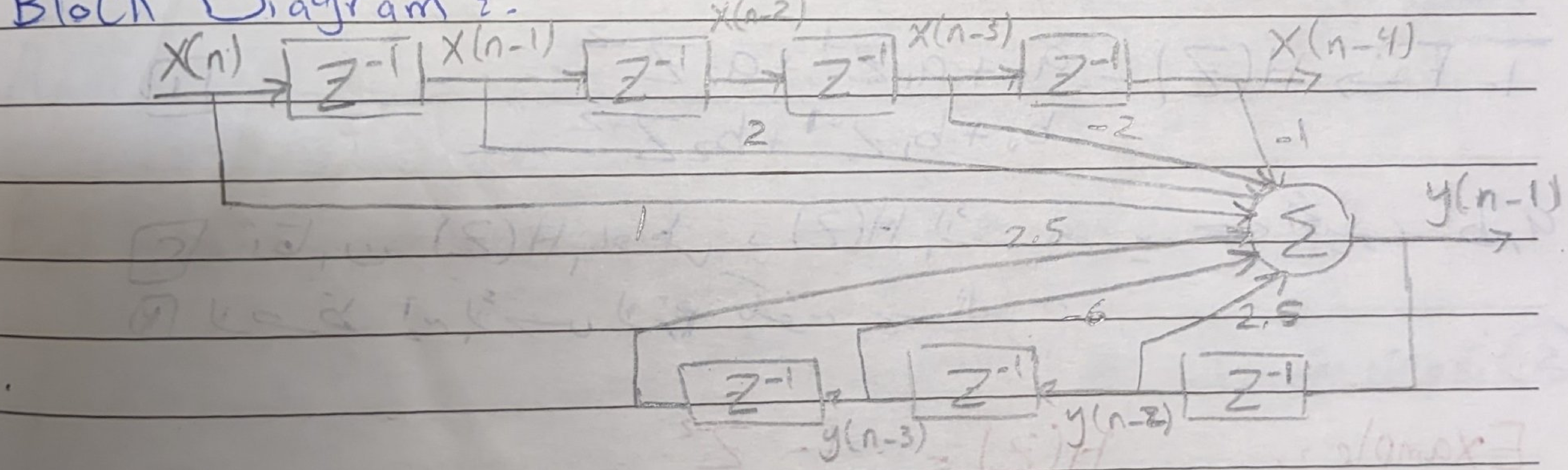
هنا هنفتكر إيه $X(Z)Z^{-k} = X(n-k)$

$$\rightarrow y(n-1) - 2.5y(n-2) + 6y(n-3) - 2.5y(n-4)$$
$$= X(n) + 2X(n-1) - 2X(n-3) - X(n-4)$$

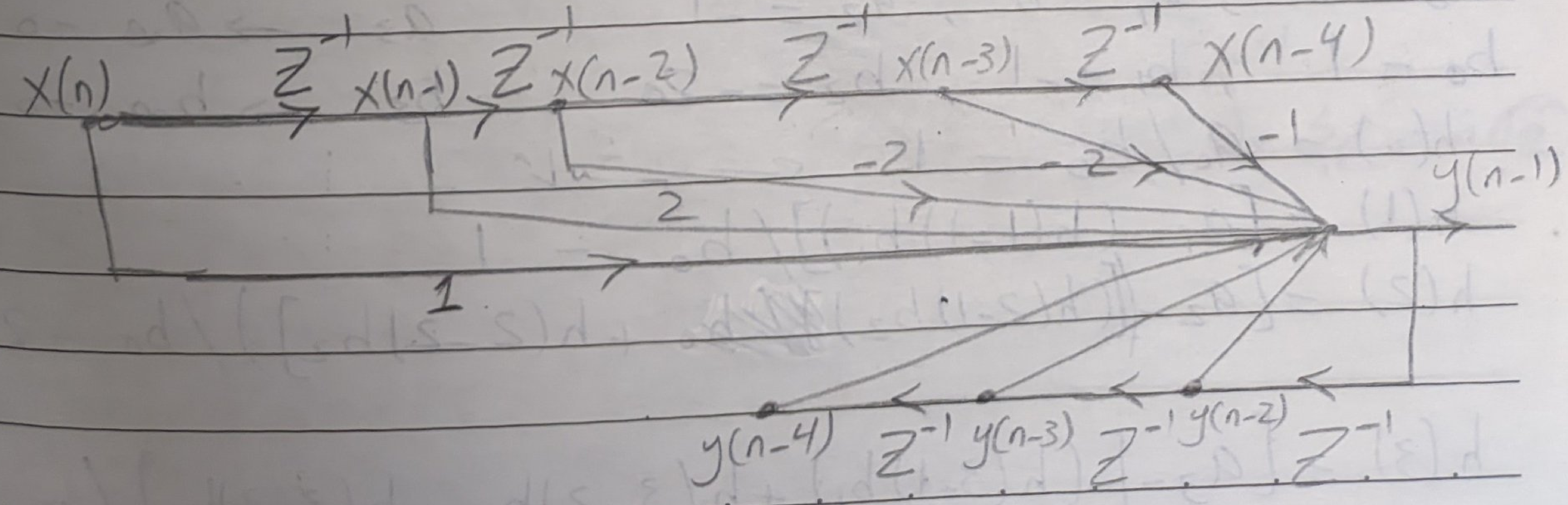
المفروض $y(n)$ من طرف لكن مفيش هاخد $y(n-1)$

$$\rightarrow y(n-1) = X(n) + 2X(n-1) - 2X(n-3) - X(n-4)$$
$$+ 2.5y(n-2) - 6y(n-3) + 2.5y(n-4)$$

Block Diagram :-



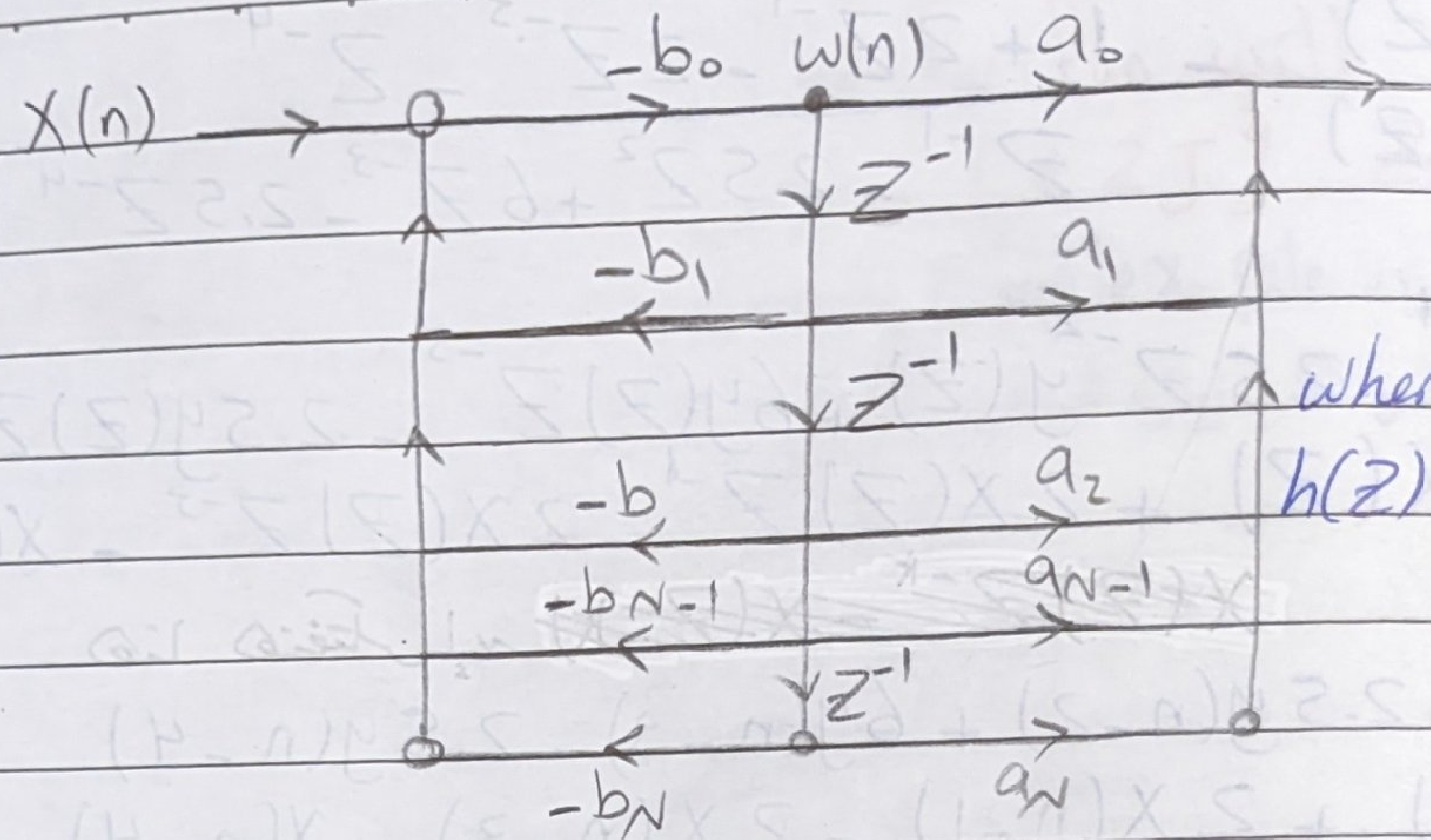
[2] Signal flow :-



Scanned by TapScanner

Date : / / التاريخ : / /

Subject : موضوع الدرس:



where :

$$h(Z) = \frac{a_0 + a_1 Z^{-1} + \dots}{b_0 + b_1 Z^{-1} + b_2 Z^{-2}}$$

→ Inverse of Z-Transform :

$\boxed{1}$ power series :

Impulse response $h(n) = [a_n - \sum_{i=1}^{n} h(n-1)\, b_i] / b_0$

$$T.F \Rightarrow H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-2} - - - -}{b_0 + b_1 Z^{-1} + b_2 Z^{-2} - - -} \quad \boxed{1}$$

$\boxed{ع}$ نقارن H(Z) المعطى بـ H(Z) الأساسية ⟸ أجيب a, b

$\boxed{٢}$ لاحظ إن الأس لازم يكون سالب

Example :
$$H(Z) = \frac{1 - Z^{-2}}{1 - Z^{-1} - 20 Z^{-2}}$$

$a_0 = 1 \quad , a_1 = 0 \quad , a_2 = -1 \qquad a_3 \longrightarrow a_\infty = 0$

$b_0 = 1 \quad , b_1 = -1 \quad , b_2 = -20 \qquad b_3 \longrightarrow b_\infty = 0$

$h(0) = a_0 / b_0 = 1$ ⟸ ثابت

$h(1) = [a_1 - (h(1-1)\, b_1)] / b_0 = 1$

$h(2) = [a_2 - ([(h(2-1)\, b_2) \quad + h(2-2)\, b_2]]) / b_0 = 20$

$h(3) = [a_3 - [(h(3-1)\, b_1) + h(3-2)\, b_2 + h(3-3)\, b_3] / b_0 = 40$

Date: / /　　　　　　　　　　　　　　　　　　التاريخ: / /

Subject:　　　　　　　　　　　　　　　　　　موضوع الدرس:

$\boxed{2}$ partial fraction:-

Ex:

$$H(Z) = \frac{Z^2}{Z^2 - 2Z - 8} \quad \text{find } x(n) \text{ using p.f}$$

① نقسم الطرفين (ع Z

$$\frac{H(Z)}{Z} = \frac{Z^{\cancel{2}}}{\cancel{Z}(Z^2 - 2Z - 8)}$$

$$= \frac{Z}{(Z+2)(Z-4)}$$

$$\longrightarrow \frac{X(Z)}{Z} = \frac{A}{Z+2} + \frac{B}{Z-4} \quad ②$$

③ نوجد قيم A ، B

$$A \text{ at } Z = -2 = \frac{Z}{Z-4} = \frac{-2}{-2-4} = \frac{1}{3}$$

$$B_{\text{at}}\ Z = 4 = \frac{Z}{Z+2} = \frac{4}{4+2} = \frac{2}{3}$$

④ نعوض بقيم الـ A ، B في المعادلة الأساسية

$$\longrightarrow \frac{X(Z)}{Z} = \frac{1/3}{Z+2} + \frac{2/3}{Z-4}$$

$$\longrightarrow X(Z) = \frac{(1/3)Z}{Z+2} + \frac{(2/3)Z}{Z-4}$$

⑤ نبص ع الجدول (مهم*) [بيكون معطى في الامتحان] ونقارن

$x(n)$		$X(Z)$
$K\delta(n)$	$\rightleftharpoons$	K
$K(\alpha)^n$	$\rightleftharpoons$	$(KZ)/(Z-\alpha)$
$Kn(\alpha)^n$	$\rightleftharpoons$	$(K\alpha Z)/(Z-\alpha)^2$

Scanned by TapScanner

$\longrightarrow \therefore X(n) = \frac{2}{3}(4)^n + \frac{1}{3}(-2)^n$

وممكن يطلب مني قيم 5 sample مثلاً من $(n=0 \rightarrow n=4)$

Example: $X(Z) = \frac{Z^2-1}{(Z-3)(Z+2)^2}$

$$\frac{X(Z)}{Z} = \frac{Z^2-1}{Z(Z-3)(Z+2)^2}$$

$$= \frac{A}{Z} + \frac{B}{(Z-3)} + \frac{C_2}{(Z+2)^2} + \frac{C_1}{(Z+2)}$$

$$A = \frac{Z^2-1}{(Z-3)(Z+2)^2}\Bigg|_{Z=0} = \frac{-1}{(-3)(2)^2} = \frac{1}{12} \quad \cancel{=}$$

$$B = \frac{Z^2-1}{Z(Z+2)^2}\Bigg|_{Z=3} = \frac{9-1}{3(3+2)^2} = \frac{8}{75} \quad \cancel{=}$$

$$C_2 = \frac{Z^2-1}{Z(Z-3)}\Bigg|_{Z=-2} = \frac{4-1}{-2(-2-3)} = \frac{3}{10}$$

$$C_1 = \frac{d}{dz}\left(\frac{Z^2-1}{Z(Z-3)}\right)\Bigg|_{Z=-2}$$

* تفاضل البسط x المقام – تفاضل المقام x البسط / (المقام)^2

$$C_1 = \frac{2Z(Z^2-3Z)-(2Z-3)(Z^2-1)}{(Z^2-3Z)^2}\Bigg|_{Z=-2} = \frac{-19}{100}$$

$$\frac{X(Z)}{\cancel{Z}} = \frac{\overset{k}{1/12\cancel{Z}}}{\cancel{Z}} + \frac{(8/75)Z}{Z-3} + \frac{(3/10)Z}{(Z+2)^2} - \frac{(19/100)Z}{Z+2}$$

$$\longrightarrow \frac{K\alpha Z}{(Z-\alpha)^2}$$

Scanned by TapScanner

$X(k) = K\,\delta(n) + \frac{8}{75}(3)^n + \left(\frac{-19}{100}\right)(-2)^n + \frac{3}{20}(-2)^n n$

To get values of 4_ samples :.

$X(0) = \frac{1}{12}(\delta(0)) + \frac{8}{75} - \frac{19}{100} =$

$\boxed{3}$ Residue :.

$$\boxed{F(Z) = X(Z)\,Z^{n-1}}$$

Example :. $X(Z) = \frac{Z^2}{Z^2-2Z-8}$ find $X(n)$ using Residue

$F(Z) = \frac{Z^2 \cdot Z^{n-1}}{Z^2-2Z-8}$ $\boxed{1}$

$= \frac{Z^{n+1}}{Z^2-2Z-8} = \frac{Z^{n+1}}{(Z-4)(Z+2)}$

$X(n) = \text{Res}\{F(Z), 4\} + \text{Res}\{F(Z), -2\}$ $\boxed{٢}$

Res نوصيل ال ٣

$\text{Res}\{F(Z), 4\} = \left.\frac{Z^{n+1}}{Z+2}\right|_{Z=4}$

$= \frac{(4)^{n+1}}{6} = \frac{2}{3}(4)^n$

$\text{Res}\{F(Z), -2\} = \left.\frac{Z^{n+1}}{Z-4}\right|_{Z=-2}$

$= \frac{(-2)^{n+1}}{-6} = \frac{1}{3}(-2)^n$

$X(n) = \frac{1}{3}(-2)^n + \frac{2}{3}(4)^n$

Scanned by TapScanner

Date : / /　　　　التاريخ : / /

Subject :　　　　موضوع الدرس:

Example $\boxed{2}$ Find $x(n)$ $\quad X(Z) = \dfrac{Z^2 - 1}{(Z-5)^2(Z+4)}$

Residue Method:

$$X(Z) = \frac{(Z^2-1)\,Z^{n-1}}{(Z-5)^2(Z+4)}$$

$$x(n) = \text{res}[F(Z), 5] + \text{res}[F(Z), -4]$$

$$\text{Res}[F(Z),5] = \frac{d}{dz}\frac{(Z^2-1)\,Z^{n-1}}{(Z+4)}\Bigg|_{Z=5} =$$

$$\frac{\left((n+1)Z^n - (n-1)Z^{n-2}\right)(Z+4) - (1)\left(Z^{n+1} - Z^{n-1}\right)}{(Z+4)^2}\Bigg|_{Z=5}$$

تفاضل البسط x المقام - تفاضل المقام x البسط

(المقام)^2

$$\frac{\left(n(5)^n + (5)^n - n(5)^{n-2} + (5)^{n-2}\right)(9) - \left(5^{n+1} - 5^{n-1}\right)}{81}$$

$$\frac{\left[9n(5)^n + 9(5)^n - \frac{9n(5^n)}{5^2} + \frac{9(5)^n}{5^2} - (5^n)(5) + \frac{5^n}{5}\right]}{81}$$

$$= \frac{9}{81}n(5)^n + \frac{9}{81}(5)^n - \frac{9*81}{25}(5)^n - \frac{81}{5}5^n$$

$$\text{Res}[F(Z), -4] = \frac{(Z^2-1)Z^{n-1}}{(Z-5)^2}\Bigg|_{-4}$$

$$\frac{15}{81}\cdot(-4)^{n-1} = \frac{15}{81}\cdot\frac{(-4)^n}{-4} = \frac{-15}{324}(-4)^n$$

$$x(n) = \frac{-15}{324}(-4)^n + \frac{9}{81}n(5)^n + \frac{9}{81}(5)^n - \frac{729}{25}(5)^n - \frac{81}{5}5^n$$

Scanned by TapScanner

(ده خاص بهم) آخر شابتر

Ch (5)

Digital Filters

FIR	IIR
Finite Impulse Response	Infinite Impulse Response
$H(Z) = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$ oR $= a_0 + a_1 Z^{-1} + a_2 Z^{-2}$ 2nd order	$H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-2}}{1 + b_1 Z^{-1} + b_2 Z^{-2}}$ - There is poles
- Found Realization structure (block Diagram) with direct method + Lattic	- Found Relazation structure (block diagram) with direct method only.

⟹ For FIR:

- Differenco equation from Transfer function:

$$H(Z) = \frac{y(Z)}{x(Z)} = h(0) + h(1)Z^{-1} + h(2)Z^{-2}$$

$$\to y(Z) = h(0)X(Z) + h(1)X(Z)Z^{-1} + h(2)X(Z)Z^{-2}$$

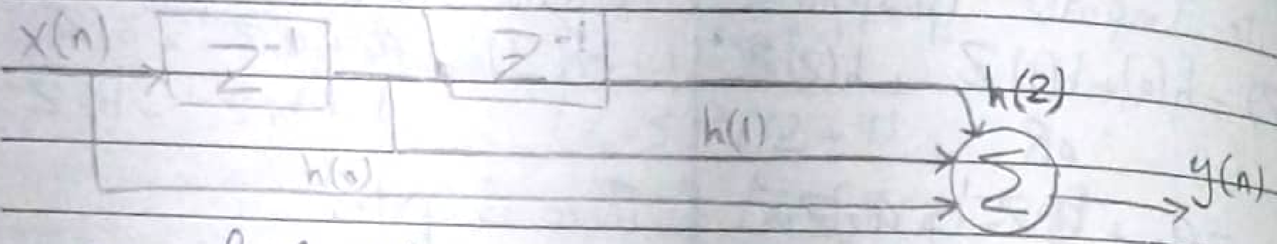
→ Difference eq.:

$$y(n) = h(0)X(n) + h(1)X(n-1) + h(2)X(n-2)$$

Date: / /

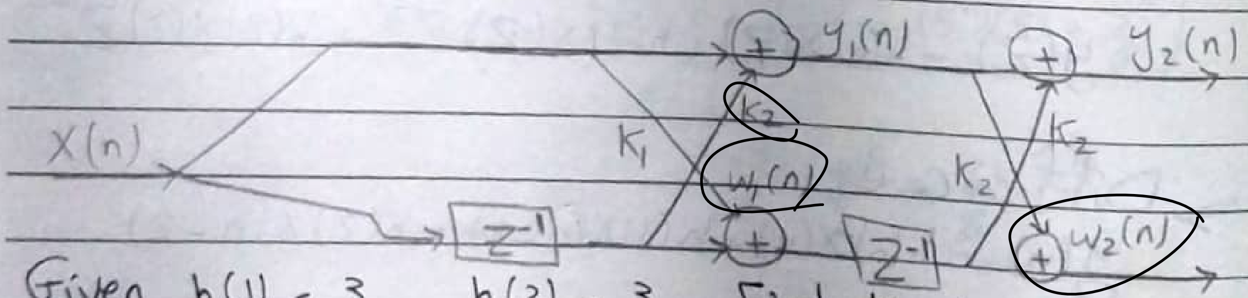
Subject:

→ block Diagram (Direct method) Transversal



- No of Addition = Input to Summer − 1
  = 3 − 1 = 2
- No of Mull = 3
- storage = I/P + Delay + Coeff
  2 + 2 + 3 = 7

→ Lattice Methods.



Given h(1) = 3 , h(2) = 3 Find K_1, K_2

From lattice : $y_1(n) = x(n) + K_1 x(n-1)$ → ①

→ $w_1(n) = K_1 x(n) + x(n-1)$ → ②

$y_2(n) = y_1(n) + K_2 w_1(n-1)$ → ③

From ② ⇒ $w_1(n-1) = K_1 x(n-1) + x(n-2)$ → ④

From (3,4) ⇒ $y_2(n) = x(n) + K_1 x(n-1) + K_1K_2 x(n-1) + K_2 x(n-$

$$y_2(n) = x(n) + \underbrace{(K_1 + K_1K_2)}_{h_1}(x(n-1)) + \underbrace{K_2}_{h_2} x(n-2) \rightarrow I$$

Difference eq for FIR:

$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2)$ II

Comparing: I & II

$\Rightarrow h(1) = K_1 + K_1K_2$

$h(2) = K_2 = 3$ $\boxed{K_2 = 3}$

$\longrightarrow h(1) = 3 = K_1 + 3K_2$

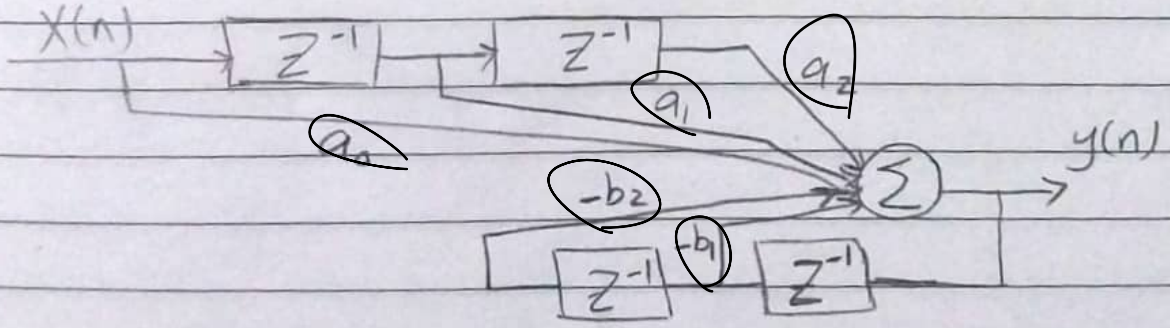
$\Rightarrow 3 = K_1 + 3 * 3$ $\boxed{K_1 = -6}$

⟹ IIR Transversal Method (Direct method)

$$H(Z) = \frac{a_0 + a_1Z^{-1} + a_2Z^{-2}}{1 + b_1Z^{-1} + b_2Z^{-2}} = \frac{y(Z)}{x(Z)}$$

$$\longrightarrow y(Z)[1 + b_1Z^{-1} + b_2Z^{-2}] = X(Z)[a_0 + a_1Z^{-1} + a_2Z^{-2}$$

$$y(n) = a_0x(n) + a_1x(n-1) + a_2x(n-2) - b_1y(n-1) - b_2y(n-2)$$



No of Addition = 4   No of Mult = 5

storage = 2 + 4 + 5 = 11

Compare between FIR & IIR:

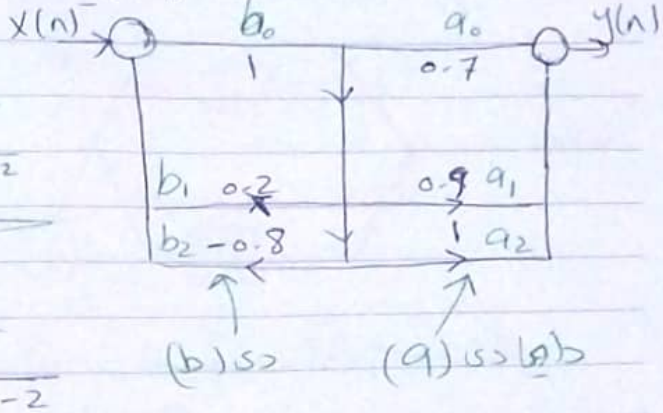
View of Comparison	FIR	IIR
No of Mult = No of Coeff	3 h(0) — h(2)	5 $a_0 \rightarrow a_2$ $b_1 \rightarrow b_2$
No of Addition - I/p to Sum - 1	3 - 1 = 2	5 - 1 = 4
Storage Memory	3 Coeff + 2 shift register + 2 I/P - 7	5 Coeff + 2 shift + 2 I/P - 9

Find Two-step difference equation relating [y(n) and x(n)]

مثال IIR وده معناه إني هشتغل بطريقة واحدة بس

هي طريقة Direct method إني أجيب H(Z)

طالما IIR يبقى بالمعادلة دي بالشكل العام



$$H(Z) = \frac{a_0 + a_1 Z^{-1} + a_2 Z^{-2}}{b_0 + b_1 Z^{-1} + b_2 Z^{-2}}$$

$$\downarrow$$

$$H(Z) = \frac{0.7 + 0.9 Z^{-1} + Z^{-2}}{1 + 0.2 Z^{-1} - 0.8 Z^{-2}}$$

دي ال Transfer function بس هو عاوز y(n)

بيقى أوصلها بالطريقة المعروفة

$$\frac{y(Z)}{X(Z)} = \frac{0.7 + 0.9 Z^{-1} + Z^{-2}}{1 + 0.2 Z^{-1} - 0.8 Z^{-2}}$$

طرفين في وسطين

$$y(Z) + 0.2 y(Z) Z^{-1} - 0.8 y(Z) Z^{-2} = 0.7 X(Z) + 0.9 X(Z) Z^{-1} + X(Z) Z^{-2}$$

طبعاً هنا هفتكر إيه

$$y(Z) Z^{-k} \longrightarrow y(n-k)$$

$$X(Z) Z^{-k} \longrightarrow X(n-k)$$

$$\Rightarrow y(n) + 0.2 y(n-1) - 0.8 y(n-2) = 0.7 X(n) + 0.9 X(n-1) + X(n-2)$$

هجيب ال y(n) في طرف لوحدها

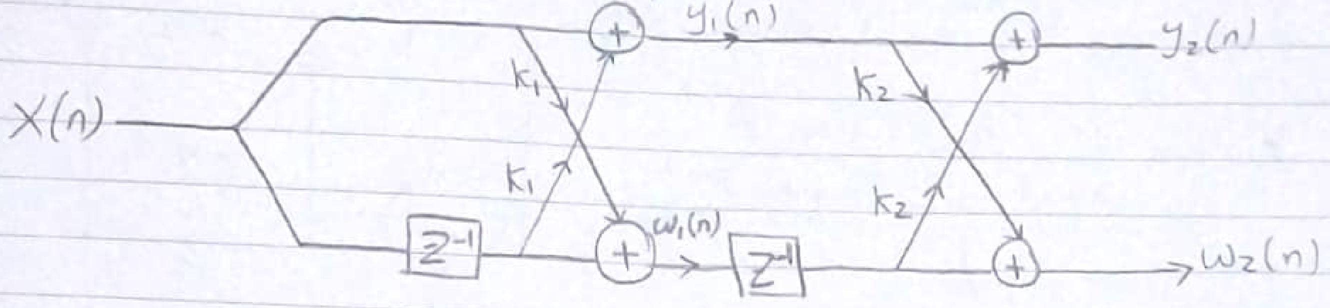
$$\Rightarrow \boxed{y(n) = 0.7 X(n) + 0.9 X(n-1) + X(n-2) - 0.2 y(n-1) + 0.8 y(n-2)} \rightarrow \text{I}$$

(هنحتاجها في المطلوب الأخير)

≠ [y(n)] ده أول مطلوب

(ii) Draw equivalent lattice

احنا هنا قولنا نحفظ الرسمة بتاعته وهرسمها



دى الرسمة الصح من الكتاب هيكون من أى رسمة تانية
وهستنتج منها المعادلات

$$y_1(n) = X(n) + K_1 X(n-1) \rightarrow ①$$

$$w_1(n) = X(n-1) + K_1 X(n) \rightarrow ②$$

$$y_2(n) = y_1(n) + K_2 w_1(n-1) \rightarrow ③$$

From ② $\rightarrow w_1(n-1) = X(n-2) + K_1 X(n-1)$

(هستنتج الـ $w_1(n-1)$ على delay بس)

وهروح في معادلة (3) أعوض عنها وعن $y_1(n)$ من ①

$$\rightarrow y_2(n) = \overbrace{X(n) + K_1 X(n-1)}^{y_1(n)} + K_2\overbrace{\left[X(n-2) + K_1 X(n-1)\right]}^{w_1(n-1)}$$

$$\rightarrow y_2(n) = X(n) + K_1 X(n-1) + K_2 X(n-2) + K_1 K_2 X(n-1)$$

هجمعهم مع بعض

$$\boxed{\rightarrow y_2(n) = X(n) + (K_1 + K_1 K_2) X(n-1) + K_2 X(n-2)} \rightarrow (\text{II})$$

(iii) Coefficients طالب نجيب الـ
اللي هما الـ K_1, K_2

بمقارنة I بـ II بالمعاملات
يعني معامل $X(n-1)$ ومعامل $X(n-2)$

تعالو نكتبهم تحت بعضا تاني ونشوفهم

I $y(n) = 0.7\,x(n) + \boxed{0.9}\,x(n-1) + \boxed{1}\,x(n-2) - 0.2\,y(n-1) + 0.8\,y(n-2)$

II $y_2(n) = x(n) + \boxed{(K_1 + K_1K_2)}\,x(n-1) + \boxed{K_2}\,x(n-2)$

$$\longrightarrow \boxed{K_2 = 1} \longrightarrow \#$$

$$K_1 + K_1\overset{1}{\cancel{K_2}} = 0.9$$

~~$K_2 + K_2$~~

$$K_1 + K_1 = 0.9$$

$$2K_1 = 0.9 \qquad K_1 = \frac{0.9}{2}$$

$$\boxed{K_1 = \frac{9}{20}} \longrightarrow \#$$